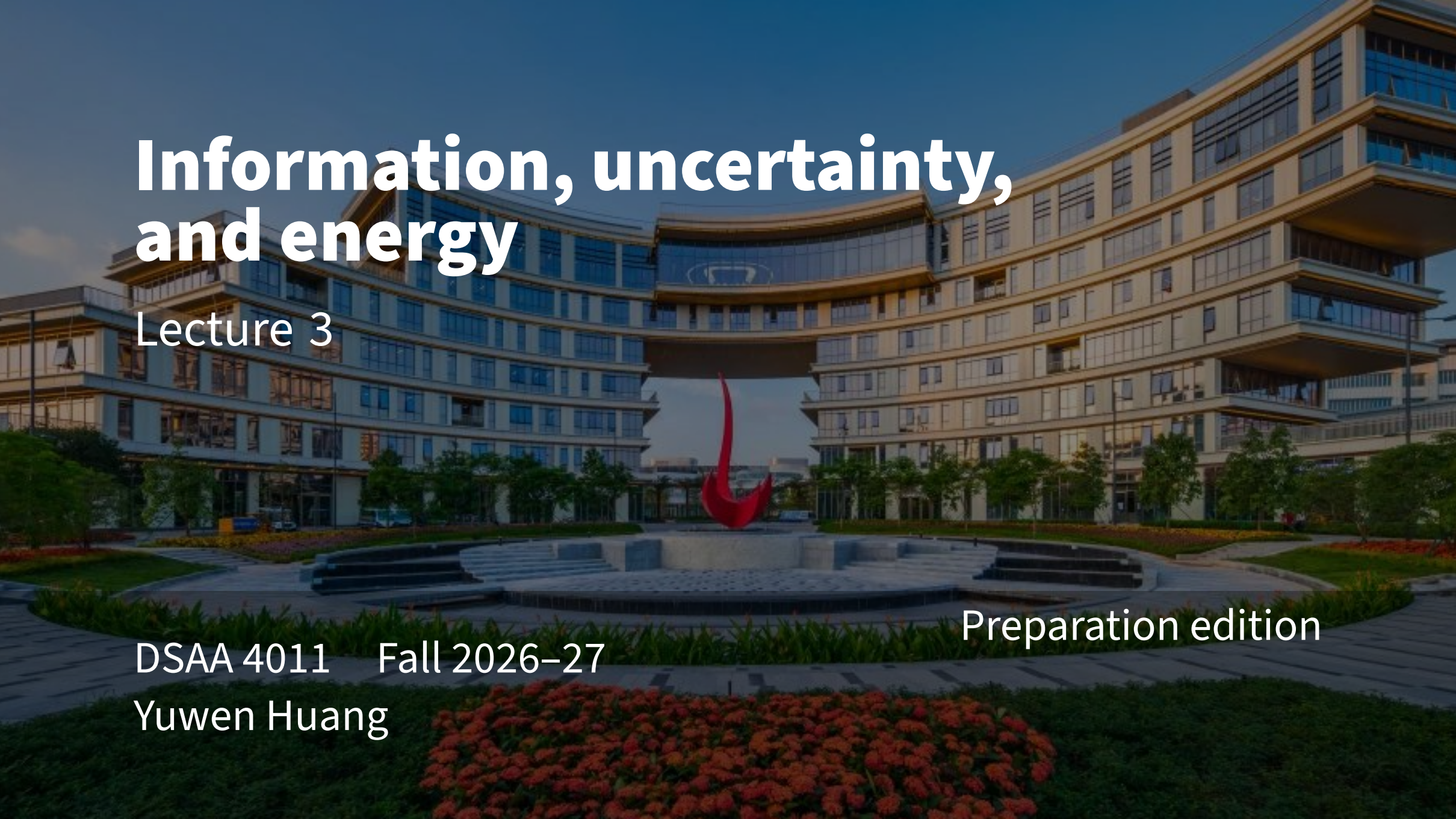


Information, uncertainty, and energy

Lecture 3

Preparation edition

DSAA 4011 Fall 2026–27

Yuwen Huang

A fitted model still needs evidence

1. Evaluate predictions
2. Measure information
3. Model with energy

Week 2: probabilities from local functions. **Now:** does the resulting model capture the system?

Predict: score probabilities against records. **Observe:** identify information that a model may miss.

A fitted model still needs evidence

1. Evaluate predictions
2. Measure information
3. Model with energy

Week 2: probabilities from local functions. **Now:** does the resulting model capture the system?

Predict: score probabilities against records. **Observe:** identify information that a model may miss.

Model: connect local energies to global behavior, then test conditions outside the fitted observations. We begin with ten records and two candidate models.

Ten observed outcomes

Observed labels

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 \end{pmatrix}$$

Six zeros and four ones.

Let $Y \in \{0, 1\}$ be a class label.
These ten outcomes form a **teaching dataset**.

A model assigns **probabilities** $q(0), q(1)$ and predicts the more probable label.

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Let $Y \in \{0, 1\}$ be a class label.
These ten outcomes form a **teaching dataset**.

A model assigns **probabilities** $q(0), q(1)$ and predicts the more probable label.

We will compare two models on these records: first their **predicted labels**, then the probabilities behind those labels.

What the two models get right

Worked example: the same ten observed labels.

Model	$q(0)$	$q(1)$	Predicted label
A	0.60	0.40	0
B	0.99	0.01	0

Six zeros, four ones.

Accuracy: fraction of correct labels.

What the two models get right

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Both choose 0 on every record. Both therefore achieve $6/10 = 60\%$ accuracy.

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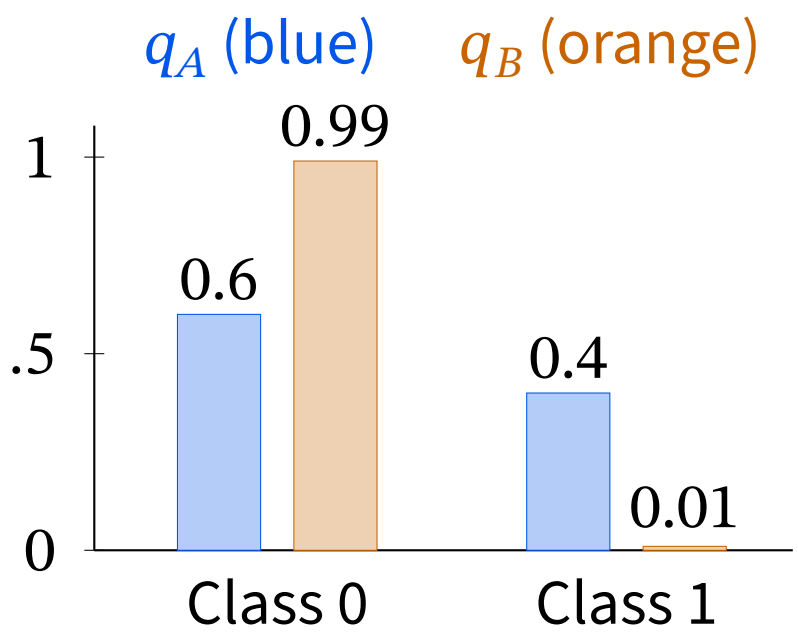
Six zeros, four ones.

Accuracy: fraction of correct labels.

Both choose 0 on every record. Both therefore achieve $6/10 = 60\%$ accuracy.

Missing information: B assigns only 1% probability to class 1, which occurs four times. Accuracy ignores this discrepancy.

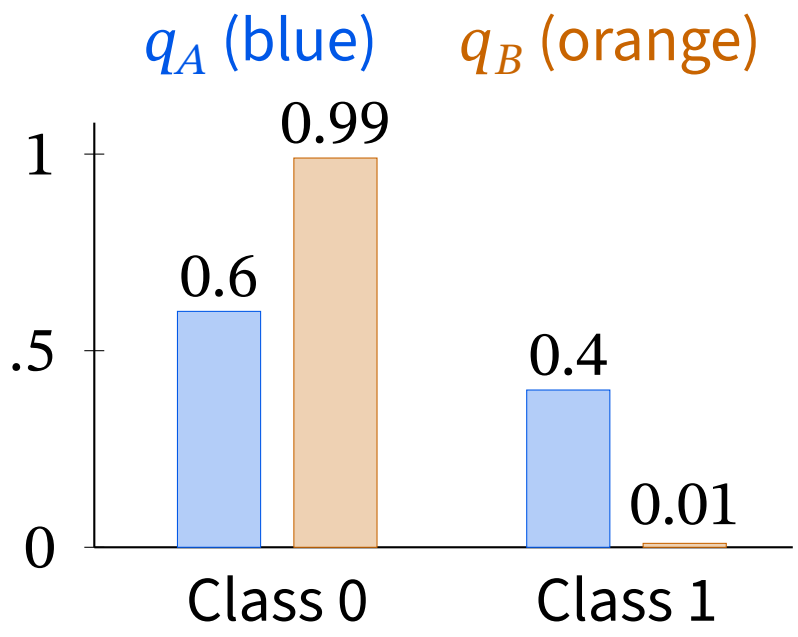
Same label, different probabilities



Accuracy could not separate A from B, so we look at their **probabilities**. Each model's $q(0), q(1)$ sum to one.

Model B assigns probability 0.01 to class 1, although class 1 occurs four times.

Same label, different probabilities



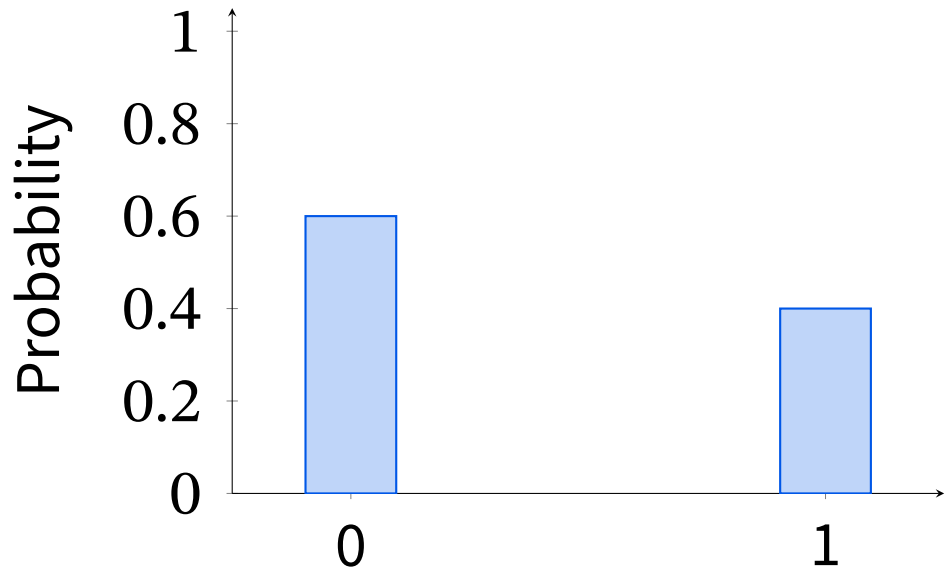
Accuracy could not separate A from B, so we look at their **probabilities**. Each model's $q(0), q(1)$ sum to one.

Model B assigns probability 0.01 to class 1, although class 1 occurs four times.

Activity 1. Which fits better: $q_C = (0.7, 0.3)$ or $q_D = (0.95, 0.05)$?

Your explanation.

A value and a distribution



Before we score probabilities, we fix notation. Y: random variable. $y \in \mathcal{Y} = \{0, 1\}$: one possible value.

The **probability mass function** satisfies $p_Y(y) \geq 0$ and

$$\sum_{y \in \mathcal{Y}} p_Y(y) = 1$$

A subscript names the variable. We omit it when the context identifies that variable.

An expectation is a weighted average

Value y / probability

$$\begin{pmatrix} 0 & 0.6 \\ 1 & 0.4 \end{pmatrix}$$

Weights sum to one.

Every score in this lecture is an average, so we define it once. For $r : \mathcal{Y} \to \mathbb{R}$,

$$\mathbb{E}_p[r(Y)] = \sum_y p_Y(y) r(y).$$

If $r(y) = y$, then

$$\mathbb{E}_p[Y] = 0(0.6) + 1(0.4) = 0.4.$$

No individual label equals 0.4. Next we ask how uncertain a distribution is.

Different distributions, different uncertainty

Blue: outcome 0

Orange: outcome 1

Certain



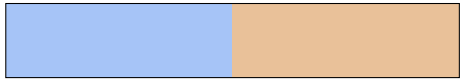
$p(0) = 1$

Biased



$p(0) = .9$

Uniform



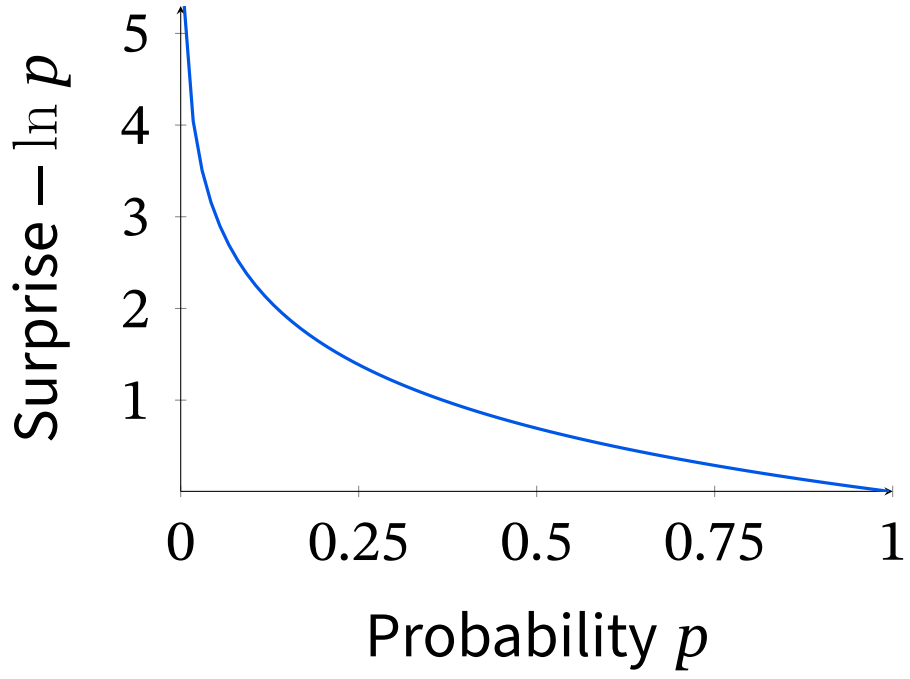
$p(0) = .5$

Return to the two-label models. All three rows describe the same two labels, yet they differ in how sure they are.

The first row leaves no uncertainty under its model. The last gives neither outcome an advantage.

We need a numerical measure that uses the **whole distribution**.

A rare outcome carries more surprise



We build that measure one outcome at a time. For $p(y) > 0$, define the surprise

$$h_p(y) = -\ln p(y).$$

Probability 1 gives surprise 0.
Probability 0.1 gives surprise 2.3026.

This quantity describes the outcome relative to p. A different model assigns a different surprise.

Why a logarithm appears

Independent outcomes

$$\begin{pmatrix} p(a) = 1/2 \\ p(b) = 1/4 \\ p(a,b) = 1/8 \end{pmatrix}$$

Independence is an assumption.

Why a logarithm of p? For independent outcomes, **probabilities** multiply:

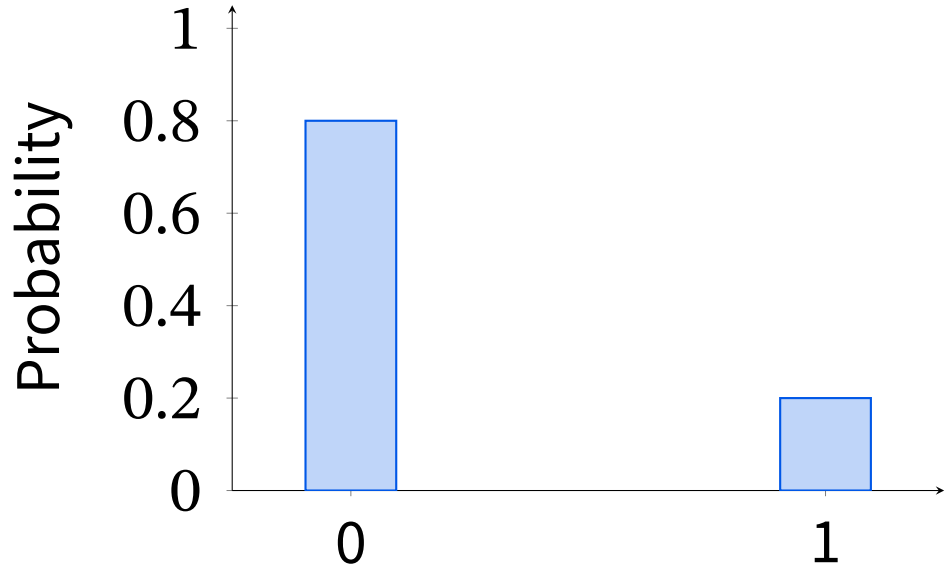
$$p(a,b) = p(a)p(b).$$

Their log surprises add:

$$-\ln[p(a)p(b)] = -\ln p(a) - \ln p(b).$$

The logarithm converts multiplication into addition. Independence must be checked before using the product.

Entropy averages surprise



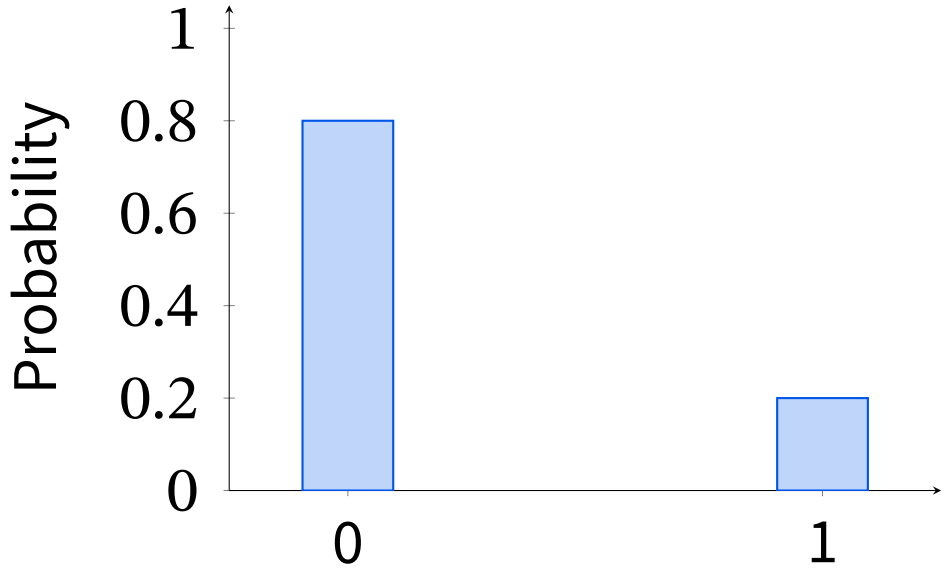
Averaging surprise over p gives its **entropy** on finite $\mathcal{Y}$:

$$H(p) = -\sum_{y:\, p(y)>0} p(y) \ln p(y).$$

For $p = (0.8, 0.2)$,

$$H(p) = 0.8(0.2231) + 0.2(1.6094).$$

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For $p = (0.8, 0.2)$,

$$H(p) = 0.8(0.2231) + 0.2(1.6094).$$

Thus $H(p) = 0.5004$ nats. The rarer outcome has greater surprise but receives a smaller averaging weight.

Two contributions to average surprise

Worked example: $p(0) = 0.8$ **and** $p(1) = 0.2$**.**

y	$p(y)$	$-\ln p(y)$	$-p(y) \ln p(y)$
0	0.8	0.2231	0.1785
1	0.2	1.6094	0.3219

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The sum is $H(p) = 0.1785 + 0.3219 = 0.5004$ **nats**.
Entropy uses probabilities as averaging weights.

Units and zero probabilities

The logarithm sets the unit

$$\begin{pmatrix} \ln 2 = 0.6931 \text{ nats} \\ \log_2 2 = 1 \text{ bit} \end{pmatrix}$$

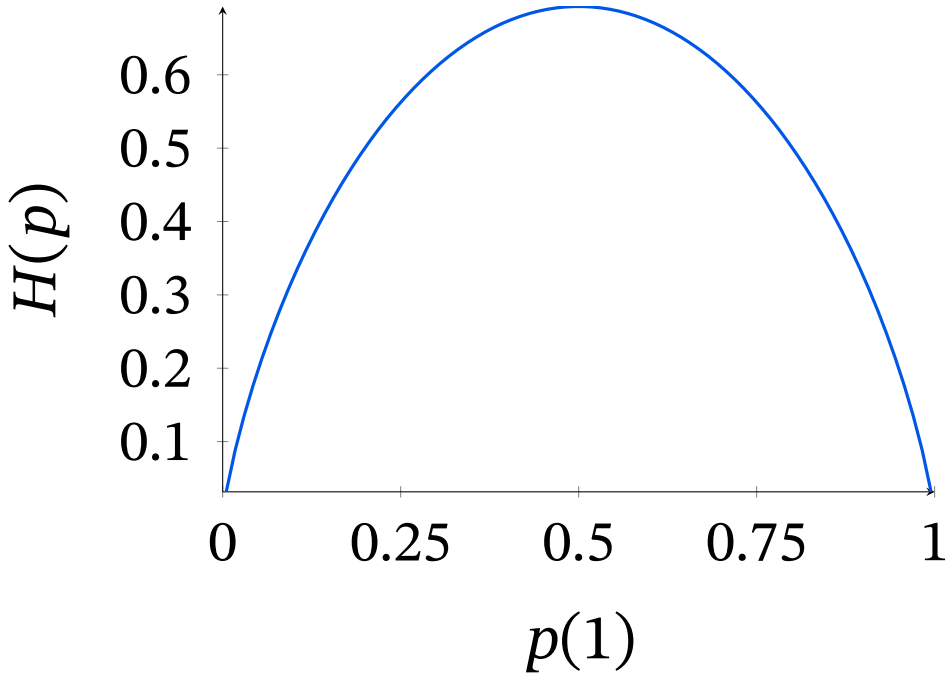
All main calculations use $\ln$.

The value 0.5004 was in nats, because we used $\ln$. Base-2 logarithms give bits. Divide nats by $\ln 2$ to obtain bits.

We use the limit convention $0 \ln 0 = 0$. An outcome with zero probability contributes zero to its model's **entropy**.

This does not permit a model to assign zero probability to an observed outcome without consequence. We return to that case in **log loss**.

Binary entropy has a maximum



Now vary the two-outcome distribution itself. For $p = (1 - t, t)$, the parameter $t \in [0, 1]$ is the probability of outcome 1.

The **entropy** is symmetric about $t = 1/2$ and vanishes at the endpoints.

Its maximum is $\ln 2$. We will prove the corresponding bound for any **finite state space** later in this lecture.

Confidence can be wrong

An outcome observed now

$$\begin{pmatrix} y = 1 \\ q(1) = 0.001 \end{pmatrix}$$

The model strongly favored 0.

At the endpoints of that curve the **entropy** is zero: the model is certain. Entropy measures only the model's own uncertainty.

If the rare label occurs, its surprise is $-\ln 0.001 = 6.9078$.

Low entropy does not certify accuracy. Evaluation against records comes in the next subsection; first, entropy for two variables.

Joint entropy keeps dependence

Two possible joint states

$$\begin{pmatrix} (0,0) : 1/2 \\ (1,1) : 1/2 \end{pmatrix}$$

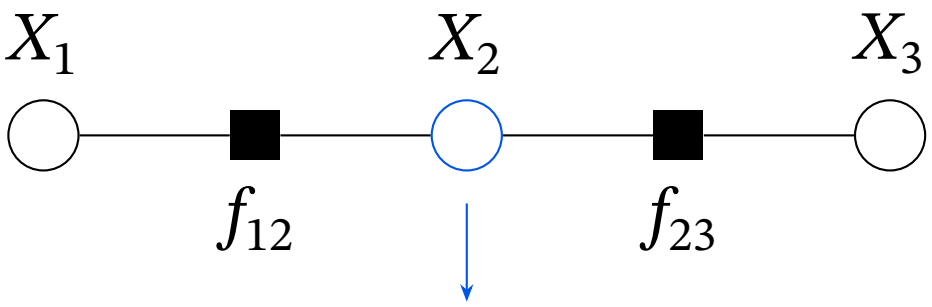
Both marginal distributions are uniform.

Let $X, Y \in \{0, 1\}$ and suppose $Y = X$. Write $H(X) = H(p_X)$. Each marginal **entropy** equals $\ln 2$.

Only two joint states occur, so $H(X, Y) = \ln 2$, not $2 \ln 2$.

Adding marginal entropies would count the **shared uncertainty twice**. We will measure this difference with mutual information.

The graph gives a distribution to study



Inference output

$p_2 = (8/13, 5/13)$

This week's question

How uncertain is X_2?

Week 2's chain, revisited.

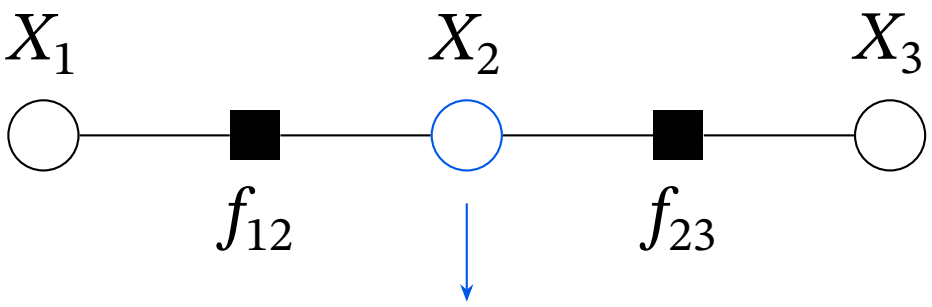
$X_i \in \{0, 1\}$ is machine i's alert state: **1 means alert**.

Rows index the first argument.

Columns index the second:

$$f_{12} = \begin{pmatrix} 4 & 1 \\ 8 & 5 \end{pmatrix}, \; f_{23} = \begin{pmatrix} 3 & 1 \\ 1 & 4 \end{pmatrix}.$$

The graph gives a distribution to study



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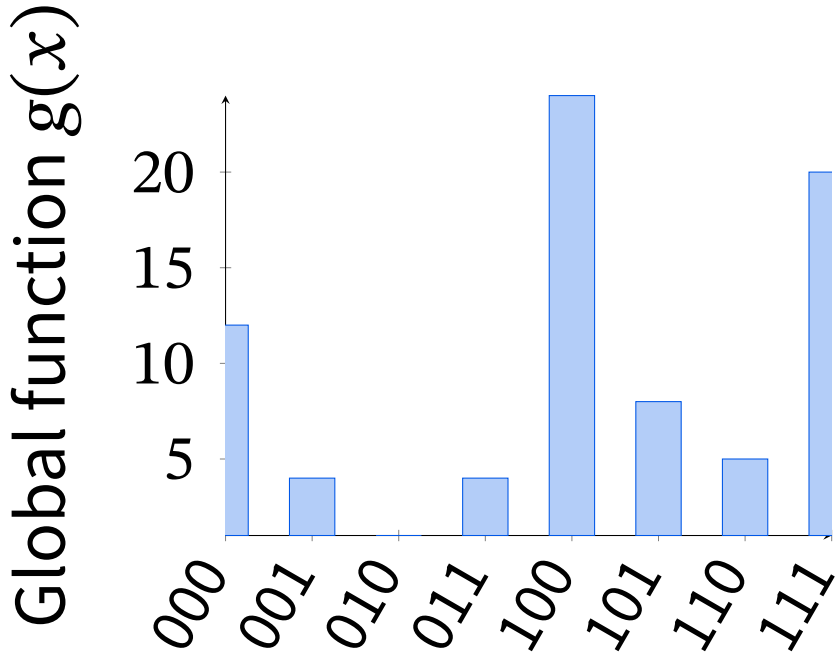
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$$f_{12} = \begin{pmatrix} 4 & 1 \\ 8 & 5 \end{pmatrix}, \; f_{23} = \begin{pmatrix} 3 & 1 \\ 1 & 4 \end{pmatrix}.$$

For $x \in \{0, 1\}^3$, the **global function** is $g(x) = f_{12}(x_1, x_2) f_{23}(x_2, x_3)$. The inferred distribution now measures uncertainty.

Eight probabilities, one normalizer



A complete state is

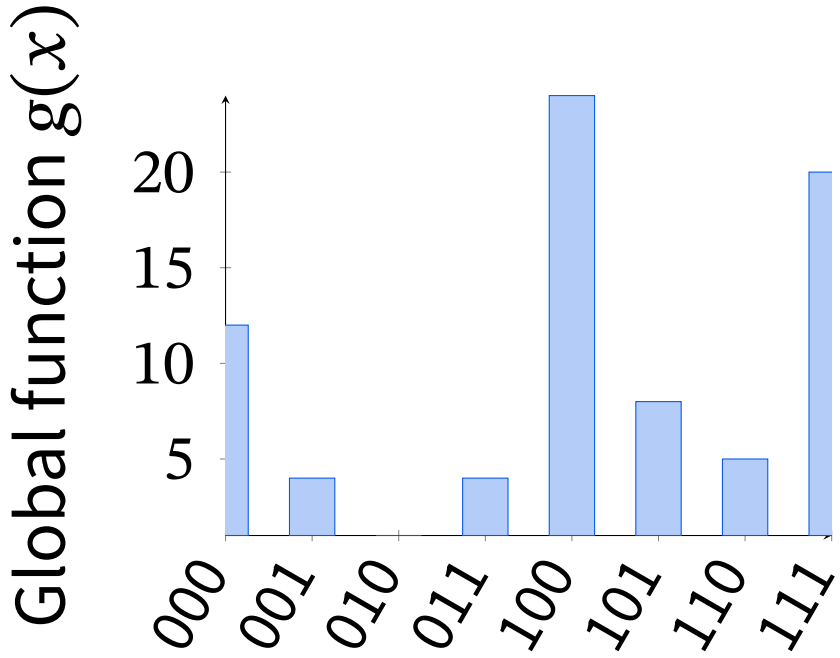
$x = (x_1, x_2, x_3) \in \mathcal{X} = \{0, 1\}^3.$

The **normalizer** is

$Z = \sum_x g(x) = 78$. In state order $000, \ldots, 111$, the probabilities are

$p = \frac{1}{78}(12, 4, 1, 4, 24, 8, 5, 20).$

Eight probabilities, one normalizer



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$p = \frac{1}{78}(12, 4, 1, 4, 24, 8, 5, 20)$.

Thus $p(x) = g(x)/78$; enumerating eight states gives $H(p) = 1.7698$ nats. **Can the local functions give this number without the full list?**

Local expectations recover joint entropy

Yes, through the factorization. Notation: $\mathcal{F}$ is the set of positive local functions. $\mathcal{N}(f)$ contains the indices of the variables read by f.

The factorization gives the log probability:

$$\ln p(x) = \sum_{f \in \mathcal{F}} \ln f(x_{\mathcal{N}(f)}) - \ln Z.$$

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$$\ln p(x) = \sum_{f \in \mathcal{F}} \ln f(x_{\mathcal{N}(f)}) - \ln Z.$$

Take the expected negative log probability:

$$H(p) = \ln Z - \sum_{f \in \mathcal{F}} \mathbb{E}_p[\ln f(X_{\mathcal{N}(f)})].$$

Linearity requires no independence assumption.

Why the marginal gives the exact expectation

Let $r : \{0,1\}^2 \to \mathbb{R}$ be any function of x_1, x_2 only.

$$\mathbb{E}_p[r(X_1, X_2)] = \sum_{x_1, x_2} \sum_{x_3} p(x_1, x_2, x_3) r(x_1, x_2).$$

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For fixed x_1, x_2, $r(x_1, x_2)$ **is unchanged** as x_3 varies:

$$\mathbb{E}_p[r(X_1, X_2)] = \sum_{x_1,x_2} r(x_1, x_2) \underbrace{\sum_{x_3} p(x_1, x_2, x_3)}_{p_{12}(x_1,x_2)}.$$

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Thus $\mathbb{E}_p[r(X_1, X_2)] = \mathbb{E}_{p_{12}}[r(X_1, X_2)]$ **exactly**: the pair marginal suffices.

Grouping states gives the pair marginal

Worked example: keep (x_1, x_2) and sum over x_3.

(x_1, x_2)	$78p(x_1, x_2, 0)$	$78p(x_1, x_2, 1)$	$78p_{12}(x_1, x_2)$
$(0, 0)$	12	4	$12 + 4 = 16$
$(0, 1)$	1	4	$1 + 4 = 5$
$(1, 0)$	24	8	$24 + 8 = 32$
$(1, 1)$	5	20	$5 + 20 = 25$

Grouping states gives the pair marginal

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$(0, 0)$	12	4	$12 + 4 = 16$
$(0, 1)$	1	4	$1 + 4 = 5$
$(1, 0)$	24	8	$24 + 8 = 32$
$(1, 1)$	5	20	$5 + 20 = 25$

For $(x_1, x_2) = (0, 0)$, **the function value stays the same**:

$$\frac{12}{78} \ln 4 + \frac{4}{78} \ln 4 = \frac{16}{78} \ln 4.$$

The probability weights add.

The first local expectation

Local function values

$$\left(f_{12} = \begin{pmatrix} 4 & 1 \\ 8 & 5 \end{pmatrix}\right)$$

Rows: $x_1 = 0, 1$. Columns: $x_2 = 0, 1$.

Exact pair probabilities

$$\left(p_{12} = \frac{1}{78}\begin{pmatrix} 16 & 5 \\ 32 & 25 \end{pmatrix}\right)$$

The same row and column order.

Set $r = \ln f_{12}$ in the proof. Match the entries of the two tables.

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The same row and column order.

Set $r = \ln f_{12}$ in the proof. Match the entries of the two tables.

$$\begin{aligned} \mathbb{E}_p[\ln f_{12}(X_1, X_2)] &= \frac{16 \ln 4 + 5 \ln 1 + 32 \ln 8 + 25 \ln 5}{78} \\ &= 1.6533178 \ldots . \end{aligned}$$

The second local expectation

Local function values

$$\left(f_{23} = \begin{pmatrix} 3 & 1 \\ 1 & 4 \end{pmatrix}\right)$$

Rows: $x_2 = 0, 1$. Columns: $x_3 = 0, 1$.

Exact pair probabilities

$$\left(p_{23} = \frac{1}{78}\begin{pmatrix} 36 & 12 \\ 6 & 24 \end{pmatrix}\right)$$

$p_{23}(x_2, x_3) = \sum_{x_1} p(x_1, x_2, x_3)$.

This time, $\ln f_{23}$ does not depend on x_1.

The second local expectation

Local function values

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This time, $\ln f_{23}$ does not depend on x_1.

Apply the regrouping proof with x_1 summed out:

$$\mathbb{E}_p[\ln f_{23}(X_2, X_3)] = \frac{36 \ln 3 + 12 \ln 1 + 6 \ln 1 + 24 \ln 4}{78}$$
$$= 0.9336039\ldots.$$

Both terms give the joint entropy

Recall the model: $p(x) = f_{12}(x_1, x_2) f_{23}(x_2, x_3)/78.$

Taking the expected negative logarithm gives

$$\begin{aligned} H(X_1, X_2, X_3) &= \ln 78 - \mathbb{E}_p[\ln f_{12}] - \mathbb{E}_p[\ln f_{23}] \\ &= 4.3567088\ldots - 1.6533178\ldots - 0.9336039\ldots \\ &= 1.7697871\ldots \text{ nats.} \end{aligned}$$

The same entropy as the eight-state calculation. The two expectations use exact pair marginals of the **same joint model**, together with its exact Z.

We are not adding pair entropies or assuming independence.

The entropy formula holds on general graphs

The chain shape was never used: for **any finite model** with positive local functions $\mathcal{F}$ and exact marginals $p_{\mathcal{N}(f)}$,

$$H(p) = \ln Z - \sum_{f \in \mathcal{F}} \mathbb{E}_{p_{\mathcal{N}(f)}}[\ln f(X_{\mathcal{N}(f)})].$$

Example: add $f_{13}(x_1, x_3)$ to the chain, creating a cycle.

$$H(p) = \ln Z - \mathbb{E}_{p_{12}}[\ln f_{12}] - \mathbb{E}_{p_{23}}[\ln f_{23}] - \mathbb{E}_{p_{13}}[\ln f_{13}].$$

The numbers 1.6533, 0.9336 and 78 belong to our chain only.

Exactness and computational cost

Using known marginals: eight weighted log terms in our chain.

Computing marginals: for 20 binary variables, one entry is

$$p_{12}(x_1, x_2) = \sum_{x_3,\ldots,x_{20}} p(x_1, x_2, x_3, \ldots, x_{20}).$$

Direct enumeration sums $2^{18} = 262{,}144$ terms.

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Direct enumeration sums $2^{18} = 262{,}144$ terms.

Trees: message passing is efficient for small local tables.

General graphs: exact inference can be expensive, and approximate marginals do not give exact entropy. Next: evaluation against records.

Evaluating probabilities against records

Our original dataset

$$\begin{pmatrix} 6 \text{ zeros} \\ 4 \text{ ones} \end{pmatrix}$$

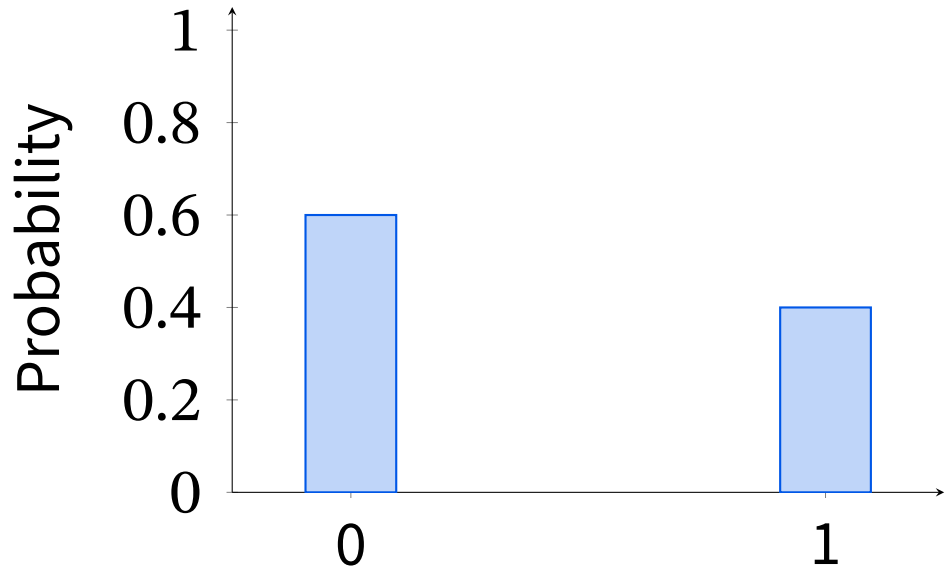
The same records evaluate both models.

We can now compute entropy, but entropy alone cannot distinguish honest uncertainty from an incorrect confident model.

For each record, inspect the probability assigned to the outcome that actually occurred.

A graphical model also predicts probabilities. **Log loss evaluates its predictions** against observations, not merely whether inference ran correctly.

A dataset defines an empirical distribution

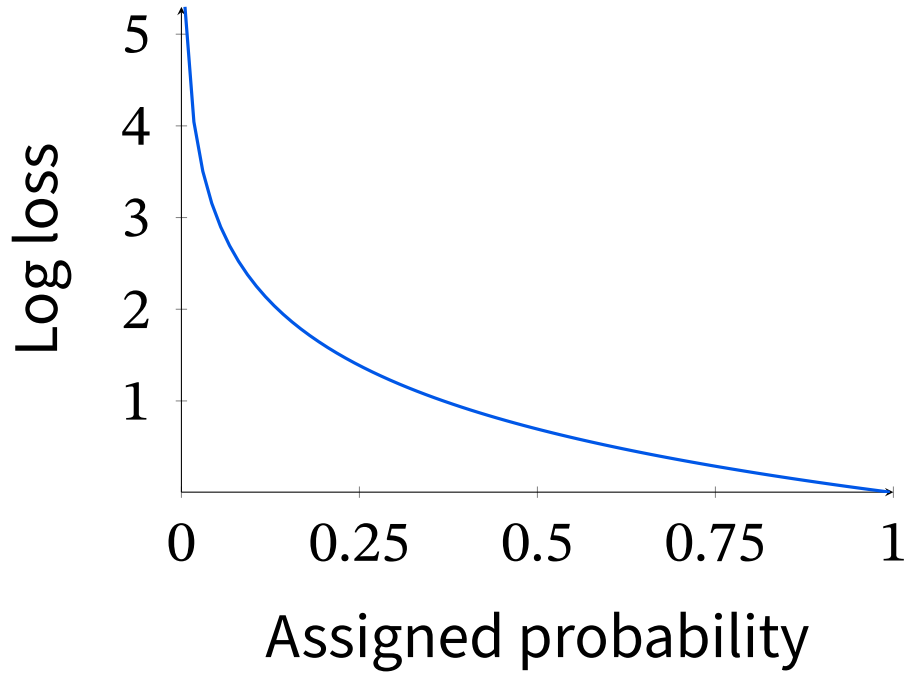


First the records themselves, as a distribution. Let $y^{(1)}, \ldots, y^{(N)}$ be N observed labels; the superscript identifies a record.

Define $\hat{p}(y)$ as the fraction of records equal to y. Here $N = 10$ and $\hat{p} = (0.6, 0.4)$.

This is the distribution of this dataset. It need not equal the population distribution.

Loss for one observed label



Now the model's side: for one record, score the probability given to what occurred. For model q and observed label y,

$$\ell(q, y) = -\ln q(y).$$

For a class-1 record, model A incurs $-\ln 0.4 = 0.9163$; model B incurs $-\ln 0.01 = 4.6052$.

An observed outcome assigned very little probability receives a large loss.

Averaging the losses

Average negative log-likelihood

$$\left(L(q) = -\frac{1}{N}\sum_{n=1}^{N} \ln q(y^{(n)})\right)$$

Smaller is better on these records.

One record gives one loss; a dataset gives their average. Group equal labels instead of writing all ten terms.

Model A:

$-0.6 \ln 0.6 - 0.4 \ln 0.4 = 0.6730.$

Model B: $-0.6 \ln 0.99 - 0.4 \ln 0.01 = 1.8481.$ This calculation answers the opening comparison without relying on accuracy alone.

The two averages, step by step

Surprise of each label

$$\begin{pmatrix} -\ln 0.6 = 0.5108 \\ -\ln 0.4 = 0.9163 \\ -\ln 0.99 = 0.0101 \\ -\ln 0.01 = 4.6052 \end{pmatrix}$$

Model A: first two rows. Model B: last two.

Weight each surprise by the label's frequency, 0.6 and 0.4, and add.

Model A:

$$0.6(0.5108) + 0.4(0.9163) = 0.3065 + 0.3665 = 0.6730.$$

Model B: $0.6(0.0101) + 0.4(4.6052) = 0.0060 + 1.8421 = 1.8481$. Four rare-label records with a large surprise dominate B's loss.

Cross-entropy separates data and model

Two roles

$$\begin{pmatrix} p: & \text{averaging distribution} \\ q: & \text{predicted distribution} \end{pmatrix}$$

Both use the same state space.

Both averages had one form: data frequencies times model surprises.

Define

$$H(p, q) = -\sum_{y} p(y) \ln q(y).$$

For the dataset, $L(q) = H(\hat{p}, q)$.

The **first argument** supplies frequencies. The **second** supplies predicted probabilities. Swapping them changes the question.

The cost of using the wrong distribution

Compare two expected losses

$$\begin{pmatrix} H(p,q) \\ H(p,p) = H(p) \end{pmatrix}$$

The observations follow p.

How much of $H(p,q)$ is due to the wrong model? Subtract the loss obtained using p itself.

Define the Kullback–Leibler (KL) divergence:

$$D_{\mathrm{KL}}(p\|q) = \sum_{y\,:\,p(y)>0} p(y) \ln \frac{p(y)}{q(y)}.$$

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If $p(y) > 0$ but $q(y) = 0$, the divergence is $+\infty$.

KL divergence is not a distance

$$p = (0.6, 0.4), q = (0.9, 0.1)$$

$$\begin{pmatrix} D_{\mathrm{KL}}(p\|q) = 0.3112 \\ D_{\mathrm{KL}}(q\|p) = 0.2263 \end{pmatrix}$$

The two directions differ.

KL divergence is **not symmetric**: $D_{\mathrm{KL}}(p\|q) \neq D_{\mathrm{KL}}(q\|p)$ in general. There is no triangle inequality either.

The first argument supplies the averaging weights, so the two directions ask different questions.

Data first: the cross-entropy loss. **Model first:** variational inference (Lecture Slides 6).

One identity connects the quantities

To link all three, add and subtract $\ln p(y)$ inside **cross-entropy**:

$$H(p, q) = -\sum_y p(y) \left[\ln p(y) + \ln \frac{q(y)}{p(y)} \right].$$

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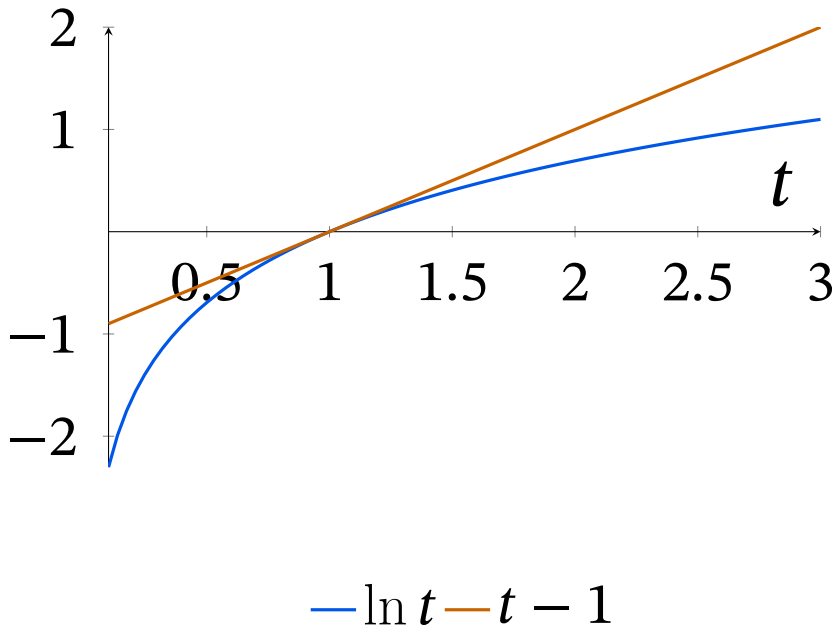
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Identify the two sums: $H(p, q) = H(p) + D_{\mathrm{KL}}(p \| q)$. For fixed p, changing q changes only the divergence.

The inequality behind KL



Can the divergence be negative? The answer rests on one inequality: for every $t > 0$, $\ln t \leq t - 1$. The curves touch at $t = 1$.

The difference $t - 1 - \ln t$ decreases before 1 and increases after 1, since its derivative is $1 - 1/t$.

Its minimum is zero. **Equality** holds only at $t = 1$.

Gibbs' inequality

Substitute $t = q(y)/p(y)$ for states with $p(y) > 0, q(y) > 0$:

$$p(y) \ln \frac{q(y)}{p(y)} \leq q(y) - p(y).$$

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Sum over the support of p:

$$-D_{\mathrm{KL}}(p\|q) \leq \sum_{y\,:\,p(y)>0} q(y) - 1 \leq 0.$$

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Sum over the support of p:

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Therefore $D_{\mathrm{KL}}(p\|q) \geq 0$. **Equality** requires $p = q$, including zero-probability states.

Detailed derivation in Further reading

Entropy bounds follow from KL

Uniform reference on K states

$$\begin{pmatrix} u(y) = 1/K \\ D_{\mathrm{KL}}(p\|u) = \ln K - H(p) \end{pmatrix}$$

Here $K = |\mathcal{Y}|$.

Nonnegativity also settles the bound promised on the binary-entropy page. Substitute the uniform reference into the definition of KL.

Nonnegativity gives $H(p) \leq \ln K$, with equality only for the uniform distribution.

Also $H(p) \geq 0$, because every nonzero surprise is nonnegative. **Equality** holds only for a certain outcome.

Which probability minimizes the loss?

A model predicts $q_t = (1 - t, t)$, where $0 < t < 1$.

The observed class-1 frequency is 0.4. Its average **log loss** is

$$L(t) = -0.6 \ln(1 - t) - 0.4 \ln t.$$

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$$L'(t) = \frac{0.6}{1 - t} - \frac{0.4}{t} = \frac{t - 0.4}{t(1 - t)}.$$

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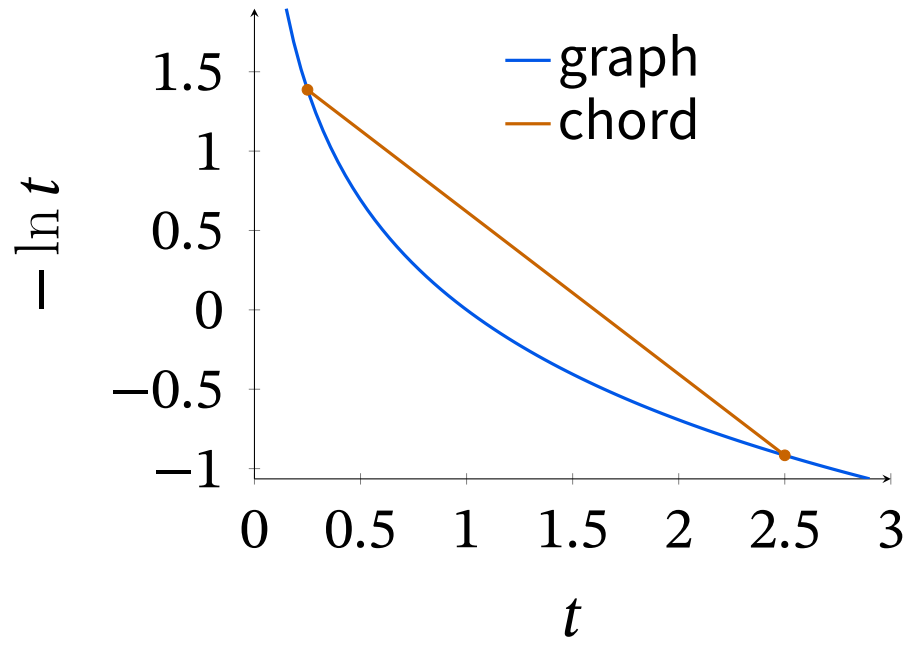
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The loss decreases before 0.4 and increases after 0.4. Thus the best constant prediction on these records is $q = (0.6, 0.4)$. This conclusion concerns the supplied data.

A convex function has no false minimum



When is a stationary point the global minimum? f is **convex** if every chord lies on or above its graph: for $0 \leq \lambda \leq 1$,

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y).$$

$-\ln t$ bends upward, so it is convex. A sum of convex functions is convex.

Any local minimum of a convex function is its global minimum.

Next: the log loss.

KL divergence is convex in the model

Fix $p = (0.6, 0.4)$

$$\begin{pmatrix} D(p\|(0.9, 0.1)) = 0.3112 \\ D(p\|(0.3, 0.7)) = 0.1920 \\ D(p\|(0.6, 0.4)) = 0 \end{pmatrix}$$

Midpoint $0 <$ average 0.2516.

Each term $-p(y) \ln q(y)$ is convex in $q(y)$, so $D_{\mathrm{KL}}(p\|q)$ and $H(p, q)$ are **convex in q**.

Its only minimum is $q = p$ (Gibbs). **Minimizing log loss over probabilities cannot stop at a false minimum.**

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Its only minimum is $q = p$ (Gibbs). **Minimizing log loss over probabilities cannot stop at a false minimum.**

Softmax layer: cross-entropy is convex in the logits and the last layer's weights, **not** in earlier layers.

Detailed derivation in Further reading

The likelihood of the whole dataset

Ten independent records

$$\begin{pmatrix} \mathcal{L}(q_A) = 0.6^6\,0.4^4 = 0.001194 \\ \mathcal{L}(q_B) = 0.99^6\,0.01^4 = 9.4 \times 10^{-9} \end{pmatrix}$$

Six zeros and four ones.

From the model's side: the **likelihood** $\mathcal{L}(q)$ is the probability q gives to the records:

$$\mathcal{L}(q) = \prod_{n=1}^{N} q(y^{(n)}).$$

Maximum likelihood picks the model that makes the records most probable: A.

Long products underflow, so we take logarithms.

Reading the factor $q(y^{(n)})$

Records 3 and 7 under model A

$$\begin{pmatrix} y^{(3)} = 0 : & q_A(y^{(3)}) = 0.6 \\ y^{(7)} = 1 : & q_A(y^{(7)}) = 0.4 \end{pmatrix}$$

One factor per record.

Before taking logarithms, read one factor. $y^{(n)}$ is the label of record n; $q(y^{(n)})$ is the model's probability of that label.

Six records have label 0 and four have label 1, so the ten factors are six 0.6's and four 0.4's: $0.6^6\, 0.4^4$.

The product changes when a record changes, not when the order of the records changes.

Log-likelihood is minus the log loss

Take $-\frac{1}{N}\ln$

$$\begin{pmatrix} -\frac{1}{10}\ln\mathcal{L}(q_A) = 0.6730 \\ -\frac{1}{10}\ln\mathcal{L}(q_B) = 1.8481 \end{pmatrix}$$

The numbers of the average-loss page.

Now take the logarithm. The log of a product is a sum of logs:

$$-\frac{1}{N}\ln\mathcal{L}(q) =$$

$$-\frac{1}{N}\sum_{n=1}^{N}\ln q(y^{(n)}) = L(q).$$

Largest likelihood = smallest log loss. **Same task.**

Grouping equal records: $L(q) = H(\hat{p}, q)$.

Likelihood and cross-entropy

Data in the first slot

$$\big(L(q) = H(\hat{p}) + D_{\mathrm{KL}}(\hat{p}\|q)\big)$$

$H(\hat{p}) = 0.6730$ does not depend on q.

The cross-entropy identity now splits the log loss. Model A: $0.6730 = 0.6730 + 0$, since $q_A = \hat{p}$. Model B: $1.8481 = 0.6730 + 1.1751$.

Maximum likelihood minimizes the divergence from the empirical distribution to the model.

A restricted family may not contain $\hat{p}$; the best fit still minimizes the same divergence.

The evaluation data can change

Worked comparison: keep both fitted models fixed.

Evaluation records	$L(q_A)$	$L(q_B)$
Six zeros, four ones	0.6730	1.8481

$L(q) = -0.6 \ln q(0) - 0.4 \ln q(1)$: average log loss in nats.

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If the next ten records are all zero, the two losses become $-\ln 0.6 = 0.5108$ and $-\ln 0.99 = 0.0101$. The ranking reverses.

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Separate fitting from evaluation. Report which records supply the frequencies. A better training score alone does not establish better prediction on new data.

An observed event cannot be ruled out

Observed labels

$$\begin{pmatrix} 0, 0, 0, 1 \\ q = (1, 0) \end{pmatrix}$$

The fourth record has probability zero.

New data can also contain a label the model ruled out. The loss of that record is $-\ln 0 = +\infty$.

The model says that this observation is impossible. This is a **support mismatch**, not ordinary uncertainty.

A smoothing rule changes the model and must be explicit. The same scoring applies to a language model's next-token prediction.

Probabilities for the next word

Teaching example: three possible tokens.

Context c: "Please close the ____."

Next token: $Y \in \mathcal{V} = \{\text{door}, \text{window}, \text{book}\}$. Recorded token: $y_* = \mathbf{window}$.

	door	window	book
Model A: $q_A(y \mid c)$	0.60	**0.30**	0.10
Model B: $q_B(y \mid c)$	0.90	**0.05**	0.05

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Model A: $q_A(y \mid c)$	0.60	**0.30**	0.10
Model B: $q_B(y \mid c)$	0.90	**0.05**	0.05

Top prediction: door. **Recorded token: window.**

Evaluate the probability assigned to **window**.

Scoring the recorded continuation

Known: both models predicted door, but the record contains window.

$$\ell_A = -\ln q_A(\text{window} \mid c) = -\ln 0.30 = 1.2040.$$

$$\ell_B = -\ln q_B(\text{window} \mid c) = -\ln 0.05 = 2.9957.$$

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Both top-token predictions miss. **A incurs less log loss** because it assigns more probability to the observed continuation.

Scope of this result: one recorded continuation evaluates one prediction. It does not establish general language ability or factual reliability.

Average loss across token positions

One position gave one loss. For N recorded token positions, c_n is the preceding context and y_n the token at position n. The model supplies $q(y \mid c_n)$.

$$L(q) = -\frac{1}{N} \sum_{n=1}^{N} \ln q(y_n \mid c_n).$$

The **chain rule** multiplies conditional probabilities. It does not assume independent tokens.

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The **chain rule** multiplies conditional probabilities. It does not assume independent tokens.

Perplexity $= \exp L(q)$, the same loss on a different scale. Report it with the records it was measured on.

Reading a perplexity

If a model spread its probability evenly over K tokens, every record would cost $-\ln(1/K) = \ln K$, so $L = \ln K$ and $\exp L = K$.

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Perplexity K means: as uncertain as a fair choice among K tokens.

Perplexity 1: every recorded token had probability 1. Model A on "window" alone: $\exp(1.2040) = 3.33$.

Calibration asks a different question

Collect every prediction in which the model said “0.7”. The model is **calibrated** if the predicted token then appears in about 70% of those cases.

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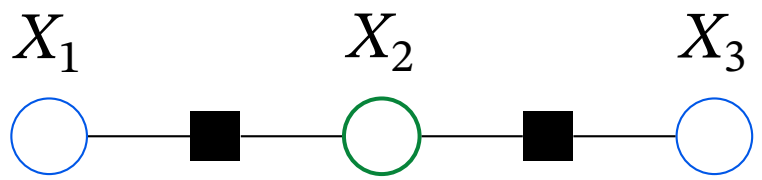
Collect every prediction in which the model said “0.7”. The model is **calibrated** if the predicted token then appears in about 70% of those cases.

Perplexity rewards probability placed on what happened. Calibration checks that stated confidence matches observed frequency. **A model can have low perplexity and still be overconfident.** Section II gives one repair: a temperature.

Break

10 minutes

Which machine should we inspect?



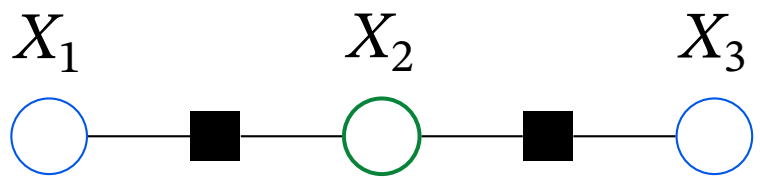
Inspect 1? Target Inspect 3?

Same graph and probabilities.
Choose an observation.

Section II asks what an observation is worth. **Model fixed.** Target: Machine 2. We may inspect Machine 1 or 3.

Each result gives a **conditional distribution** for X_2. Compare the remaining uncertainty.

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Each result gives a **conditional distribution** for X_2. Compare the remaining uncertainty.

Decision from inference: choose the observation with the greater average information gain under this model.

Conditioning selects and renormalizes

Rows x_2 / columns x_3

$$\begin{pmatrix} 36 & 12 \\ 6 & 24 \end{pmatrix}$$

Weights $b_{23} = 78 p_{23}$.

Start with inspecting Machine 3. If $X_3 = 0$, select the first column. Its total is 42.

Normalize that column: $p(X_2 \mid X_3 = 0) = (6/7, 1/7)$.

If $X_3 = 1$, normalize the second column instead: $(1/3, 2/3)$. The observation determines which column is relevant.

The remaining entropy after each result

78 $p(x_2, x_3)$, rows x_2, columns x_3

$$\begin{pmatrix} 36 & 12 \\ 6 & 24 \end{pmatrix}$$

Column sums 42 and 36.

$X_3 = 0$: column

$(36, 6)/42 = (6/7, 1/7)$;

$(6/7)\ln(7/6) + (1/7)\ln 7 =$

$0.1321 + 0.2780 = 0.4101.$

$X_3 = 1$: column

$(12, 24)/36 = (1/3, 2/3)$;

$(1/3)\ln 3 + (2/3)\ln(3/2) =$

$0.3662 + 0.2703 = 0.6365.$

Each is the entropy of X_2's conditional belief, computed exactly as $H(X_2) = 0.6663$ was.

Two possible inspection results

Target: X_2. Observation: the alert state X_3.

Observation	$p(X_3)$	$p(X_2 = 0 \mid X_3)$	Entropy
$X_3 = 0$	$42/78 = 7/13$	$36/42 = 6/7$	0.4101
$X_3 = 1$	$36/78 = 6/13$	$12/36 = 1/3$	0.6365

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Before inspection, $p(X_2) = (48/78, 30/78)$ and $H(X_2) = 0.6663$. Either result leaves less **entropy**, but we do not yet know which result will occur.

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To value the inspection **before its result is known**, weight each row by its probability $p(X_3)$ and add: the next page.

Average over possible observations

Observation / remaining entropy

$$\begin{pmatrix} X_3 = 0 : & 0.4101 \\ X_3 = 1 : & 0.6365 \end{pmatrix}$$

Observation probabilities: $7/13, 6/13$.

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The **expected remaining entropy** is

$$\frac{7}{13}(0.4101) + \frac{6}{13}(0.6365) = 0.5146.$$

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In general, conditional **entropy** averages over observations:

$$H(Y \mid X) = \sum_x p_X(x) H(p_{Y|x}).$$

Mutual information measures the reduction

Target: Machine 2

$$\begin{pmatrix} H(X_2) = 0.6663 \\ H(X_2 \mid X_3) = 0.5146 \end{pmatrix}$$

Units: nats

The inspection lowered the expected entropy. That reduction has a name: for finite X, Y, the **mutual information** (MI) is $I(X;Y) = H(Y) - H(Y \mid X)$.

Here $I(X_3;X_2) = 0.1517$ nats, using unrounded values.

Inspecting Machine 1 gives only 0.0175 nats about Machine 2 under this model. Machine 3 gives the **larger average reduction**.

Dependence compared with independence

Is the reduction always nonnegative? Compare the joint with the independent reference having the same **marginals**, $p_X(x)p_Y(y)$:

$$D_{\mathrm{KL}}(p_{XY} \| p_X p_Y) = \sum_{x,y} p(x,y) \ln \frac{p(y \mid x)}{p_Y(y)}.$$

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$$D_{\mathrm{KL}}(p_{XY} \| p_X p_Y) = H(Y) - H(Y \mid X) = I(X;Y).$$

Thus $I(X;Y) \geq 0$, with equality exactly when X and Y are independent. The joint/reference comparison also proves symmetry.

One observation can increase uncertainty

A separate two-variable example

$$\begin{pmatrix} P(Y=1) = 0.9 \\ P(X=1) = 0.1 \\ P(Y=1 \mid X=1) = 0.5 \end{pmatrix}$$

A rare observation produces a balanced belief.

On average entropy cannot rise. One result can. Choose $P(Y=1 \mid X=0) = 17/18$ so that $P(Y=1)$ stays 0.9.

Activity 2. Compare $H(Y)$ with $H(Y \mid X=1)$. Does this contradict the average inequality?

One observation can increase uncertainty

A separate two-variable example

$$\begin{pmatrix} P(Y=1) = 0.9 \\ P(X=1) = 0.1 \\ P(Y=1 \mid X=1) = 0.5 \end{pmatrix}$$

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Your explanation.

Joint entropy includes remaining uncertainty

Two stages of description

$$\begin{pmatrix} X \\ Y \mid X \end{pmatrix}$$

First X, then Y given X.

Back to averages. Use the identity $p(x, y) = p_X(x)p(y \mid x)$.

Taking the expected negative log gives

$$H(X, Y) = H(X) + H(Y \mid X).$$

Substituting $I(X; Y) = H(Y) - H(Y \mid X)$ gives $I(X; Y) = H(X) + H(Y) - H(X, Y)$: adding marginal entropies counts the shared part twice; the double count is the mutual information.

Counting shared uncertainty once

Worked example: X is a fair bit and $Y = X$.

Joint state (x, y)	(0,0)	(1,1)	Other states
Probability	$1/2$	$1/2$	0

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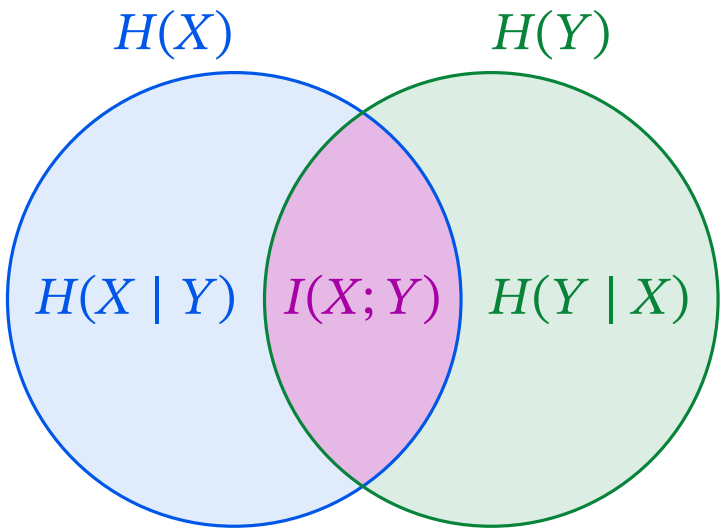
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The chain rule gives $H(X, Y) = \ln 2 + 0$. Therefore $I(X; Y) = \ln 2 + \ln 2 - \ln 2 = \ln 2$. The second variable repeats the same uncertainty.

One picture for the five quantities

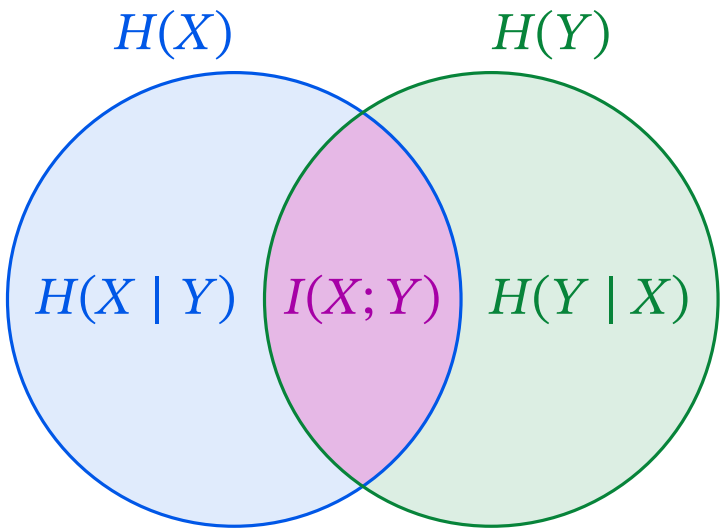


whole area: $H(X, Y)$

Any pair: each circle is a marginal entropy. Overlap: **mutual information**. Circle minus overlap: **conditional entropy** given the other.

$H(X) = H(X \mid Y) + I(X;Y)$; whole area: $H(X, Y) = H(X \mid Y) + I(X;Y) + H(Y \mid X)$.

One picture for the five quantities



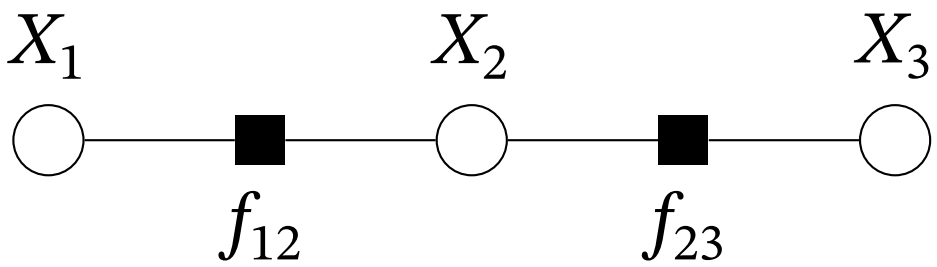
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Chain, $X = X_2$, $Y = X_3$: $H(X_2 \mid X_3) = 0.5146$, $I = 0.1517$, $H(X_3 \mid X_2) = 1.2048 - 0.6663 = 0.5385$; sum $H(X_2, X_3) = 1.2048$.

No shared local function is not independence

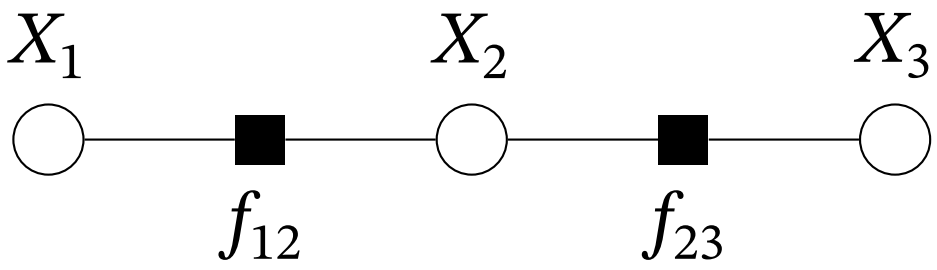


X_2 and X_3 share a local function. What about X_1 and X_3? They share no **local function**, but their joint distribution need not factorize.

In this chain $I(X_1; X_3) = 0.0049$ nats. The path through X_2 carries dependence.

After fixing X_2, the remaining two sides separate. **Conditional independence is a different statement.**

Processing cannot add missing information



That separation is the Markov property. For $X_1 \to X_2 \to X_3$, the **conditional distribution** of X_3 depends only on X_2.

Then

$$I(X_1; X_3) \leq I(X_1; X_2).$$

This requires the Markov assumption. Extra observations can still help.

Detailed derivation in Further reading

A relation invisible to pairs

Allowed states (X, Y, W)

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

Each row has probability $1/4$.

Pairwise information can also miss a relation entirely. Let X, Y be independent fair bits, and let $W = X \oplus Y$ denote their exclusive OR (XOR).

The four rows define the full **joint distribution**. We do not infer it from pairwise statistics.

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Activity 3. Replace XOR by AND: $V = XY$. Is V independent of X?

Your explanation.

Every pair looks independent

Worked example: X, Y are fair independent bits and $W = X \oplus Y$.

x	y	w	Probability
0	0	0	$1/4$
0	1	1	$1/4$
1	0	1	$1/4$
1	1	0	$1/4$

Marginal table for (X, W):

$$p_{XW} = \begin{pmatrix} 1/4 & 1/4 \\ 1/4 & 1/4 \end{pmatrix}.$$

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Not joint independence: X and Y together determine W.

Two observations reveal what one cannot

Keep the same XOR distribution.

$$H(W) = \ln 2, \qquad H(W \mid X) = H(W \mid Y) = \ln 2.$$

Seeing either input alone leaves the output fair.

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Each variable alone is a fair bit, so $H(X) + H(Y) + H(W) = 3 \ln 2$; the next board shows why the joint **entropy** is smaller.

Why the joint entropy is $2 \ln 2$, not $3 \ln 2$

Count the triples (x, y, w) that can occur.

Only four triples have positive probability, each $1/4$:
$000,\ 011,\ 101,\ 110$. A distribution uniform on four states has entropy $\ln 4 = 2 \ln 2 = 1.3863$.

Why the joint entropy is $2 \ln 2$, not $3 \ln 2$

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The chain rule says the same:
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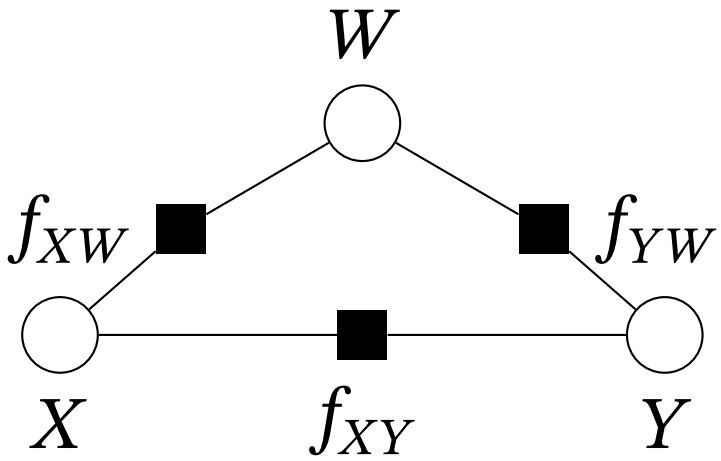
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Separately: three fair bits, $3 \ln 2 = 2.0794$. Together: two bits. **The constraint $W = X \oplus Y$ removes one bit**, invisible to any single pair.

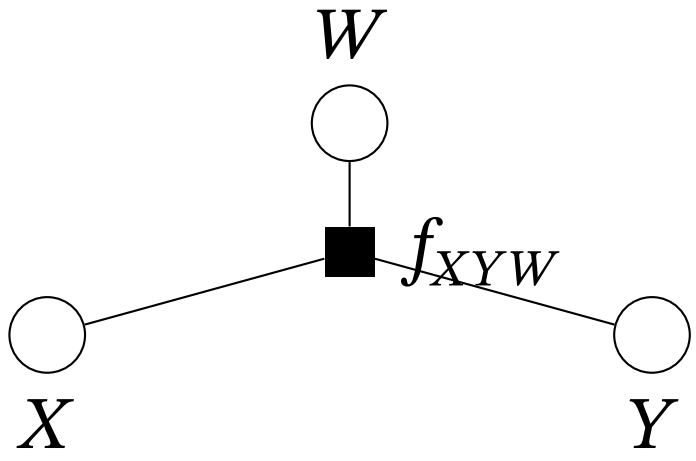
Two graphs, different distributions

Pairwise summaries



Set all three local functions to 1.

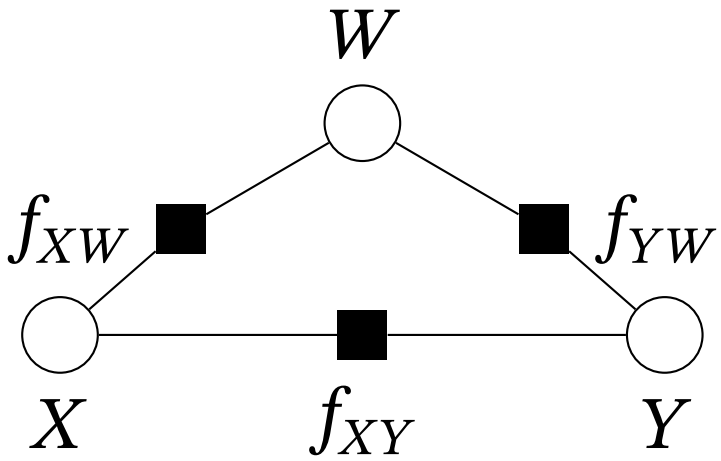
XOR constraint



$f_{XYW} = 1$ if $w = x \oplus y$.
Otherwise $f_{XYW} = 0$.

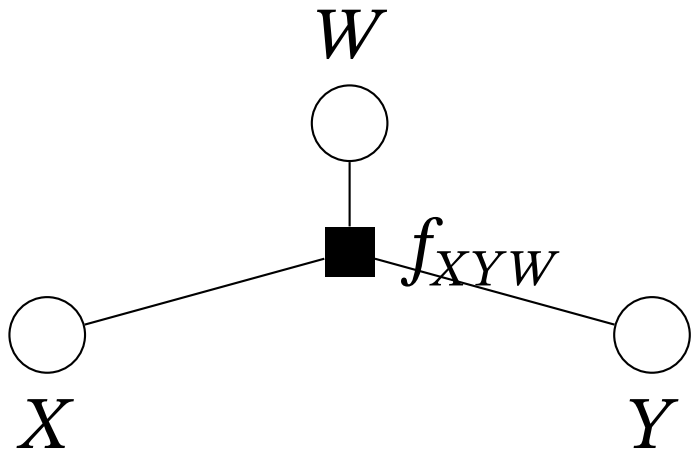
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Left: 8 states at $1/8$. **Right:** 4 allowed states at $1/4$.

Information is not causal importance

Two different questions

$$\begin{pmatrix}\text{Seeing } X\text{, what do I learn about } Y?\\ \text{Changing } X\text{, what happens to } Y?\end{pmatrix}$$

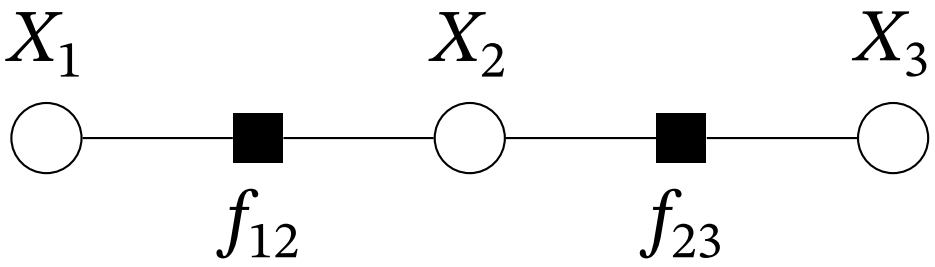
Mutual information answers only the first.

A last limit: $I(X;Y)$ is large whenever X and Y move together, whatever the reason.

Ice-cream sales and sunburn cases rise together: large MI. Ice cream causes no sunburn; sunshine drives both.

A large $I(X;Y)$ says "X is informative about Y", not "changing X changes Y". The latter needs an experiment or causal assumptions.

The same graph has an energy description

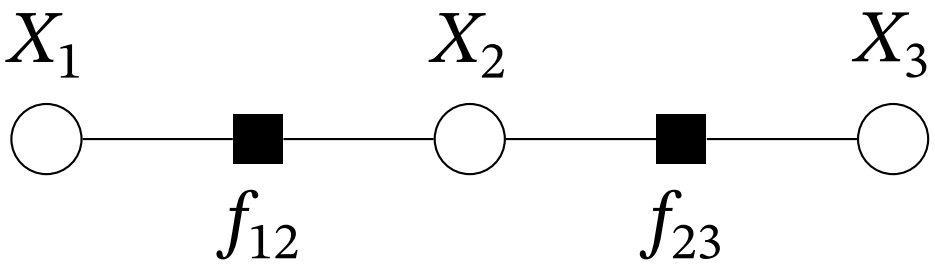


Back to the chain. Each **positive local function** weights a local combination of states.

Multiply the local functions:

$$g(x) = f_{12}(x_1, x_2) f_{23}(x_2, x_3).$$

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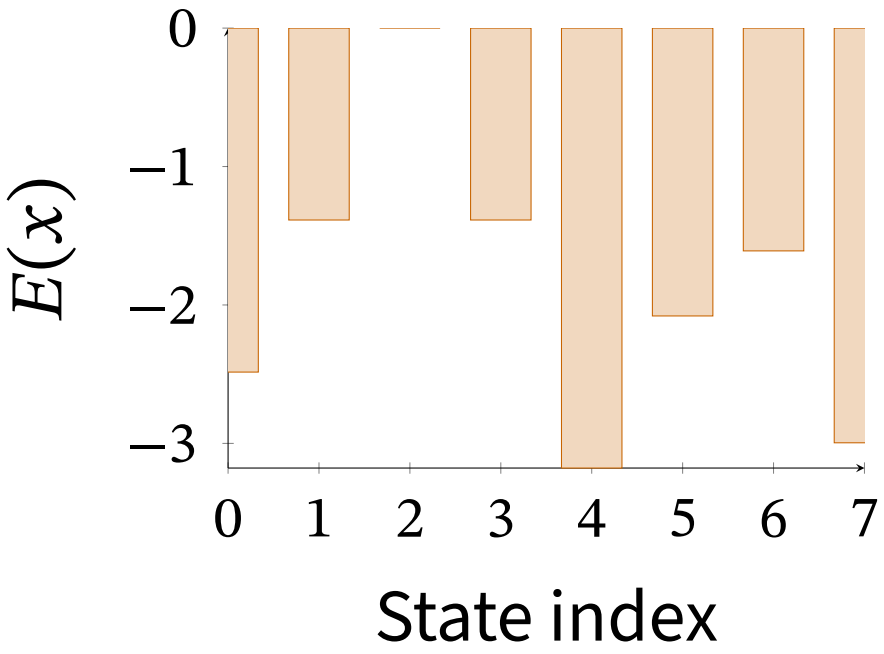
$$g(x) = f_{12}(x_1, x_2) f_{23}(x_2, x_3).$$

A negative logarithm turns the product into a sum:

$$E(x) = -\ln g(x) = -\ln f_{12}(x_1, x_2) - \ln f_{23}(x_2, x_3).$$

Why use this form? Local energy terms show which preferences cooperate or conflict.

Energy is a negative log weight



In general, define the local energy $E_f = -\ln f$ and the global energy

$$E(x) = \sum_f E_f(x_{\mathcal{N}(f)}) = -\ln g(x).$$

A weight above 1 gives a negative energy: $g = 24$ gives $E = -\ln 24 = -3.1781$.

Only **differences** matter: $p(x)/p(x') = e^{E(x')-E(x)}$. Adding one constant to every $E(x)$ changes no probability.

Why the word energy

Physics

$$\begin{pmatrix}\text{low energy} \;\Rightarrow\; \text{likely state} \\ p \propto e^{-E/(k_\mathrm{B}T)}\end{pmatrix}$$

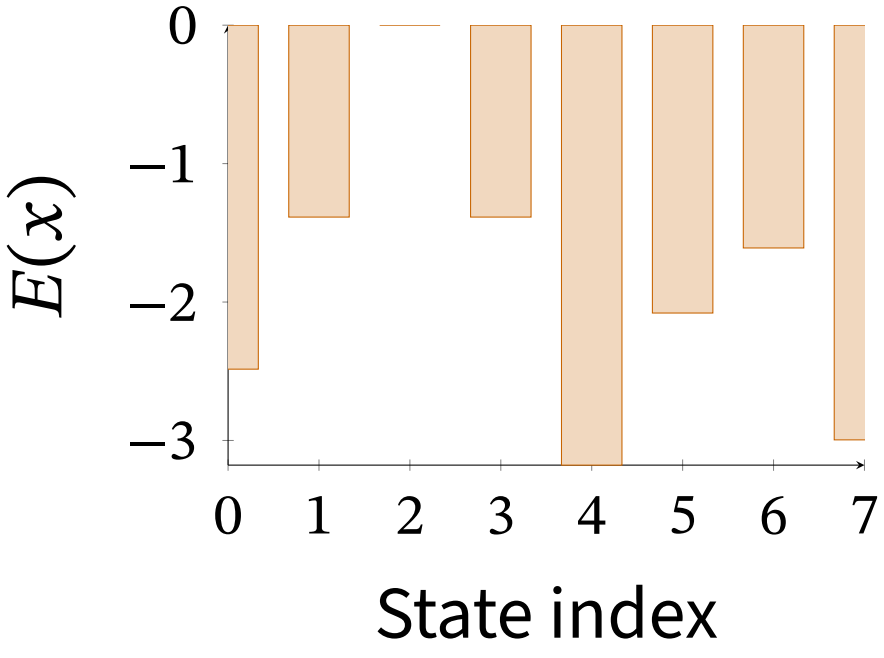
Boltzmann, 1877.

In physics a low-energy state is more probable, with probability $\propto e^{-E/(k_\mathrm{B}T)}$. Our $p = e^{-E}/Z$ has the same shape, so we borrow the word.

Here $E(x) = -\ln g(x)$ is a **score** without a unit, not a measurement in joules.

Larger energy, smaller weight: the lowest-energy state $(1, 0, 0)$ is the most probable.

A distribution from energies



Physics also divides energy by a temperature. Borrow that: for finite states and $T > 0$, define

$$Z_T = \sum_x e^{-E(x)/T}.$$

Normalize the weights:

$$p_T(x) = e^{-E(x)/T}/Z_T.$$

This is the Boltzmann distribution. At $T = 1$ it recovers the original model g/Z.

Temperature controls probability ratios

Two states a, b

$$\begin{pmatrix} E(a) = 0 \\ E(b) = 2 \end{pmatrix}$$

Only their energy difference matters.

What does T do? Compare two states; the **normalizer** cancels in their ratio:

$$\frac{p_T(b)}{p_T(a)} = e^{-2/T}.$$

At $T = 1$ the ratio is 0.1353. At $T = 2$ it is 0.3679.

Higher temperature makes this higher-energy state less strongly penalized. It does not reverse the **energy ordering**.

Temperature moves probability mass

Worked model: three states with fixed energies $E = (0, 1, 2)$.

	State 0	State 1	State 2
$E(x)$	0	1	2
$e^{-E(x)}$	1	0.3679	0.1353
$p_1(x)$	0.6652	0.2447	0.0900

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$p_1(x) = e^{-E(x)}/Z_1$ with $Z_1 = 1 + 0.3679 + 0.1353 = 1.5032$. At $T = 2$: weights $(1, 0.6065, 0.3679)$, $Z_2 = 1.9744$, $p_2 = (0.5065, 0.3072, 0.1863)$.

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Higher **temperature**: the lowest-energy state stays most probable, but the others gain.

Extreme one: very low temperature

Same three states, $E = (0, 1, 2)$. Lower the temperature.

T	$p_T(0)$	$p_T(1)$	$p_T(2)$
0.5	0.8668	0.1173	0.0159
0.1	1.0000	0.0000	0.0000

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$e^{-1/T}$ and $e^{-2/T}$ vanish next to $e^0 = 1$: all probability collapses onto the **lowest-energy state**.

m states tied at the minimum share it: $1/m$ each. In general the limit is uniform on the minimum-energy set.

Detailed derivation in Further reading

Extreme two: very high temperature

Same three states. Raise the temperature.

T	$p_T(0)$	$p_T(1)$	$p_T(2)$
10	0.3672	0.3322	0.3006
100	0.3367	0.3333	0.3300

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Every $e^{-E/T} \to 1$: the three states become **equally likely**, $1/3$ each.

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Every $e^{-E/T} \to 1$: the three states become **equally likely**, $1/3$ each.

In between, T sets how much energy differences count.

An added constant changes no probability

Shift every state equally

$$\begin{pmatrix} E'(x) = E(x) + c \\ e^{-E'(x)/T} = e^{-c/T} e^{-E(x)/T} \end{pmatrix}$$

The common multiplier cancels.

The energy page claimed that only differences matter. Verify it with Z_T present: the shifted partition function is $Z'_T = e^{-c/T} Z_T$.

Both numerator and denominator receive that factor, so $p'_T(x) = p_T(x)$.

Consequently Z_T depends on the energy baseline. A raw partition value cannot compare unrelated models without their conventions.

Softmax is the same normalization

Scores $z_1, \dots, z_K$

$$\left(q(k) = \frac{e^{z_k/T}}{\sum_j e^{z_j/T}} \right)$$

Set $E(k) = -z_k$.

Classifiers do the same. A logit is an unnormalized score for class $k \in \{1, \dots, K\}$.

Adding one constant to every logit leaves the **probabilities** unchanged.

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Temperature scaling fits T on validation records; it changes calibration, not the class ranking or accuracy.

C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger, “On calibration of modern neural networks,” ICML 2017, PMLR 70, 1321–1330.

Computing the normalizer without overflow

The softmax normalizer $Z = \sum_k e^{z_k}$ has a numerical problem: for large scores the computer cannot store even one term, an **overflow**.

The problem

$$\begin{pmatrix} z = (1000, 999) \\ e^{1000} \approx 10^{434} \end{pmatrix}$$

A double holds at most 10^{308}.

The fix: factor out the largest term. With $m = \max_k z_k$,

$$\sum_k e^{z_k} = e^m \sum_k e^{z_k - m}.$$

Every exponent $z_k - m$ is now ≤ 0, so every term lies in $(0, 1]$ and nothing overflows.

The same normalizer, computed safely

Shifted scores

$$\begin{pmatrix} z - m = (0, -1) \\ e^0 = 1,\ e^{-1} = 0.3679 \end{pmatrix}$$

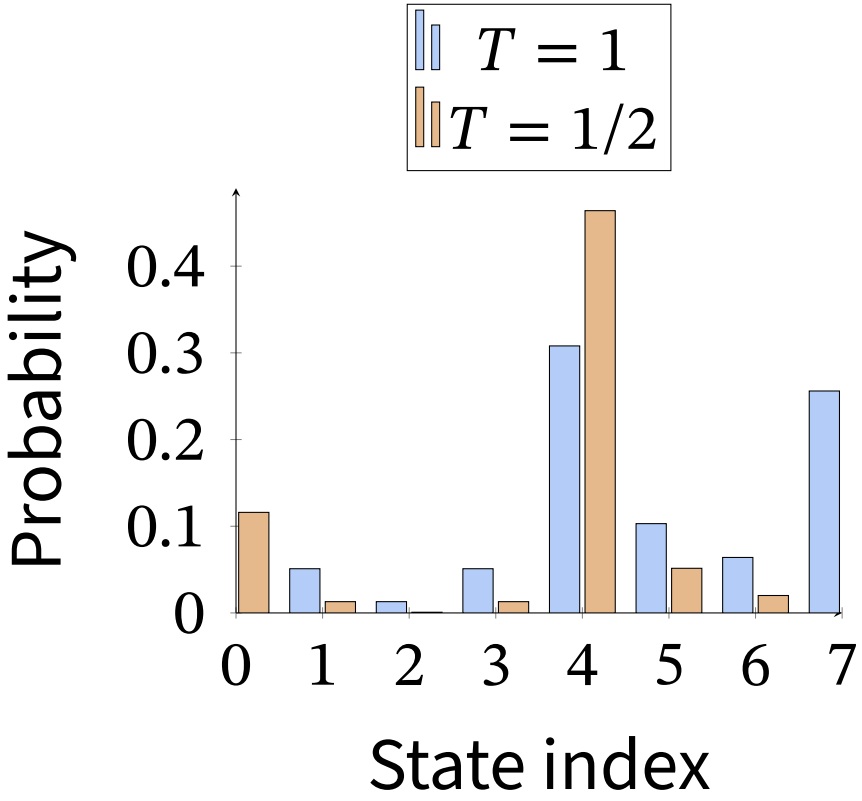
Sum 1.3679.

Logs of the factored form:
$\ln Z = m + \ln \sum_k e^{z_k - m} =$
$1000 + \ln 1.3679 = 1000.3133.$

Probabilities need only the shifted terms: $q = (1, 0.3679)/1.3679 = (0.7311, 0.2689)$.

This **log-sum-exp** routine sits inside every softmax layer and cross-entropy loss.

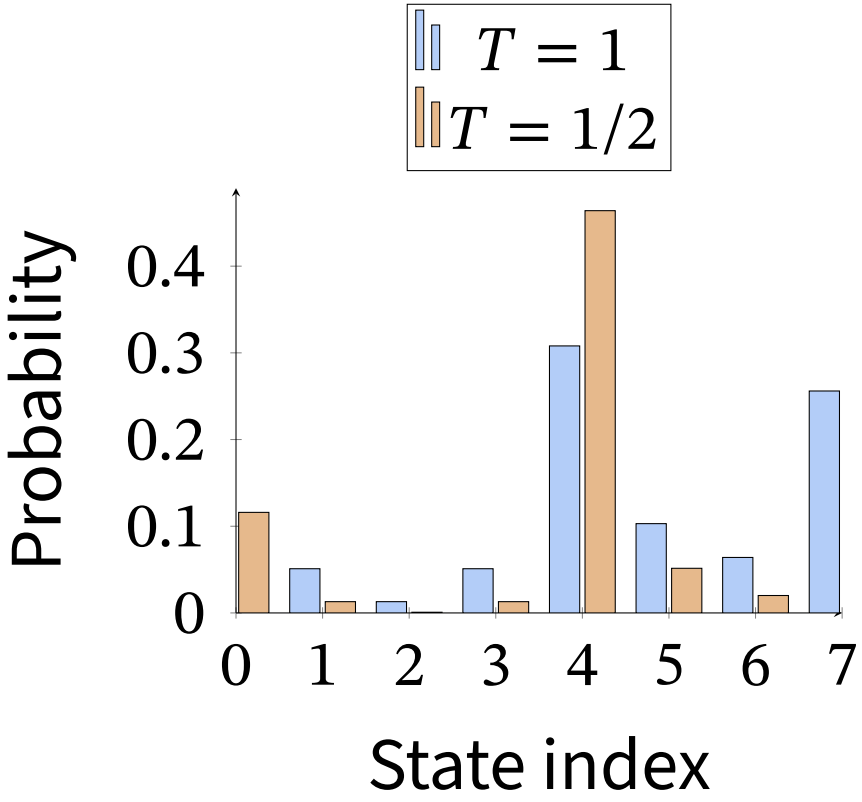
Temperature changes every local weight



Back to the chain. To **temper** it at T, raise every weight to the power $1/T$: $g \mapsto g^{1/T}$, that is $E \mapsto E/T$.

Because $g = f_{12}f_{23}$, each **local function** becomes $f^{1/T}$. At $T = 1/2$ every weight is squared; enumeration gives $Z_{1/2} = 1242$, not 78^2.

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A power of a sum and a sum of powers are different operations.

Apply the temperature before summing out

The eight weights, grouped by x_2

$$\begin{pmatrix} x_2 = 0 : \ 12,\ 4,\ 24,\ 8 \ (\text{sum } 48) \\ x_2 = 1 : \ 1,\ 4,\ 5,\ 20 \ (\text{sum } 30) \end{pmatrix}$$

Question: the belief about X_2 at $T = 1/2$?

Correct: square all eight weights, then sum the group:
$144 + 16 + 576 + 64 = 800$,
$1 + 16 + 25 + 400 = 442$;
$p_{1/2}(X_2 = 0) = 800/1242 = 0.6441.$

Wrong: square last week's marginal $(48, 30)$:
$2304/3204 = 0.7191.$

Apply the temperature before summing out

The eight weights, grouped by x_2

$$\begin{pmatrix} x_2 = 0 : \ 12,\ 4,\ 24,\ 8 \ (\text{sum } 48) \\ x_2 = 1 : \ 1,\ 4,\ 5,\ 20 \ (\text{sum } 30) \end{pmatrix}$$

Question: the belief about X_2 at $T = 1/2$?

Correct: square all eight weights, then sum the group:
$144 + 16 + 576 + 64 = 800$,
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Wrong: square last week's marginal $(48, 30)$:
$2304/3204 = 0.7191.$

$(12 + 4 + 24 + 8)^2 \neq 12^2 + 4^2 + 24^2 + 8^2$: a sum forgets how its parts were spread.

Hidden states explain the difference

Worked example: group the eight states by their value of x_2.

Group	Weights	Square, then sum	Sum, then square
$x_2 = 0$	$12, 4, 24, 8$	800	2304
$x_2 = 1$	$1, 4, 5, 20$	442	900

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$x_2 = 0$	$12, 4, 24, 8$	800	2304
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At $T = 1/2$, square each original weight before summing:

$p_T(X_2 = 0) = 800/(800 + 442) = 0.6441.$

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At $T = 1/2$, square each original weight before summing:
$p_T(X_2 = 0) = 800/(800 + 442) = 0.6441.$

Squaring the group totals instead gives $2304/(2304 + 900) = 0.7191.$
The extra cross terms correspond to different hidden configurations.
The two procedures define different distributions.

Why the order matters

Same group, two treatments

$$\begin{pmatrix} (12+4+24+8)^2 = 2304 \\ 12^2+4^2+24^2+8^2 = 800 \end{pmatrix}$$

The cross terms make the difference.

Tempering acts on each configuration's weight; a marginal has already merged those weights.

So apply the temperature to the full model, before any hidden variable is summed out.

Activity 4. Which path throws information away?
Your explanation.

Entropy and average energy

Define the average energy

$$\big(U(T) = \mathbb{E}_{p_T}[E(X)]\big)$$

The averaging distribution is p_T.

Temperature also links the two halves of this lecture, uncertainty and energy. Take a negative log:

$$-\ln p_T(x) = E(x)/T + \ln Z_T.$$

Average both sides under p_T:

$$H(p_T) = U(T)/T + \ln Z_T.$$

We use this identity in Lecture 4. Before that, a question the energy picture invites: is the lowest-energy state where the probability is?

The most probable individual state

Ten independent bits

$$\begin{pmatrix} P(X_i = 1) = 0.8 \\ x = (1,1,1,1,1,1,1,1,1,1) \end{pmatrix}$$

Each $X_i \in \{0, 1\}$.

Test this on a simple model: ten independent bits. The all-one configuration has probability $0.8^{10} = 0.1074$.

Replacing any 1 by 0 multiplies its probability by $0.2/0.8 = 1/4$.

Thus all ones is the **most probable individual configuration**. Most probability can still lie elsewhere.

Many configurations can outweigh one

One ten-one state

Two of the 45 eight-one states

1	1	1	1	1	1	1	1	1	1

1	1	1	1	1	1	1	1	0	0

0	0	1	1	1	1	1	1	1	1

A configuration with eight ones and two zeros has probability $0.8^8(0.2)^2$, below 0.8^{10}.

Their **combined probability is larger** because there are 45 of them.

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Their **combined probability is larger** because there are 45 of them.

A mode answers which individual state is most probable. Summed probability answers how likely an event is.

Why eight ones can outweigh ten

Ten independent bits satisfy $P(X_i = 1) = 0.8$.

One particular configuration with eight ones has probability $0.8^8(0.2)^2 = 0.006711$.

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Choose the two zero positions. There are $\binom{10}{2} = 10 \cdot 9/2 = 45$ choices, all with that probability. Their total is $45(0.006711) = 0.3020$.

Why eight ones can outweigh ten

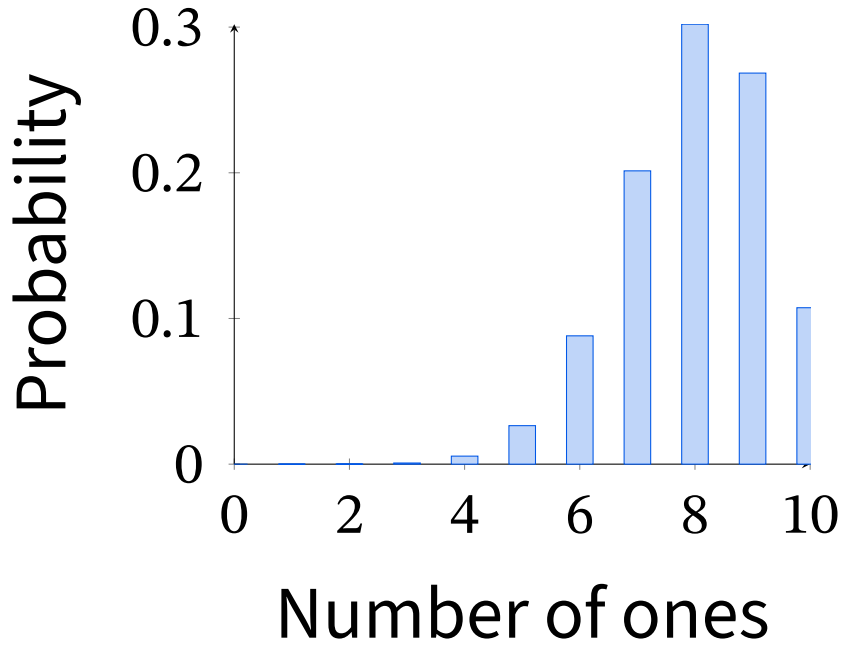
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Compare with the all-one state: $0.8^{10} = 0.1074$. Every eight-one state is individually less likely, but their **union** is more likely.

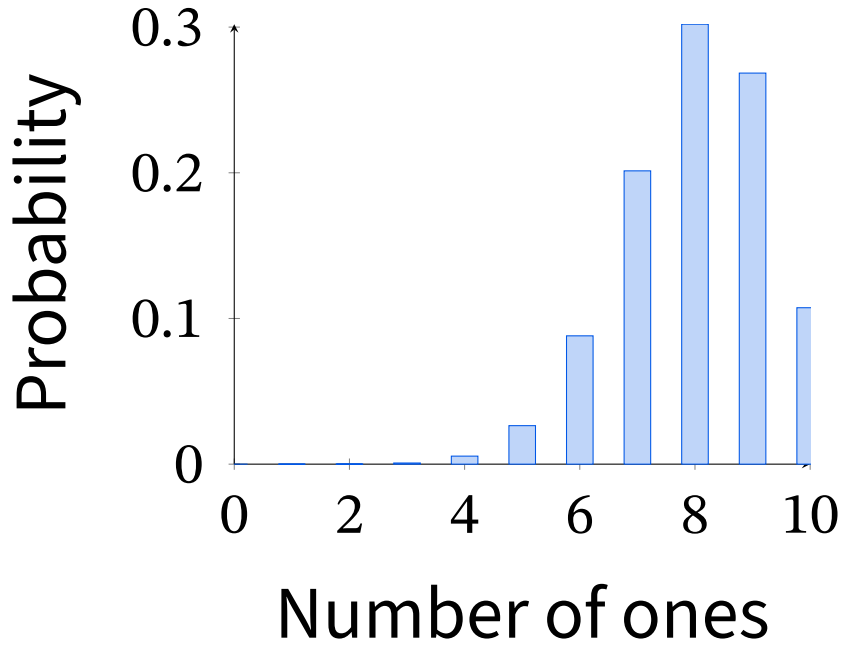
Probability grouped by number of ones



Group every configuration by its number of ones. Let $K = \sum_{i=1}^{10} X_i$ count them; its values are 0 through 10.

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Each bar sums **probabilities** of all configurations with that count.

Each bar is a count of configurations times one shared weight. We now compute the count.

Counting the configurations of each type

$$\binom{10}{k}, k = 0, \dots, 10$$

$$\begin{pmatrix} 1,\ 10,\ 45,\ 120,\ 210,\ 252, \\ 210,\ 120,\ 45,\ 10,\ 1 \end{pmatrix}$$

Sum $2^{10} = 1024$.

The **type** of a configuration is its number of ones, k. All configurations of a type share the probability $p^k(1-p)^{n-k}$.

There are $\binom{n}{k} = \dfrac{n!}{k!\,(n-k)!}$ of them, the **binomial coefficient**.

$P(K = k) = \binom{n}{k} p^k (1-p)^{n-k}$: a count times a weight.

Stirling's approximation counts for us

$$\ln 10!$$

$$\left(\begin{array}{c}\text{exact } 15.104\\ n\ln n - n = 13.026\\ +\frac{1}{2}\ln(2\pi n) = 15.096\end{array}\right)$$

Two terms suffice.

For large n the count needs an approximation. **Stirling:**

$$\ln n! \approx n\ln n - n + \tfrac{1}{2}\ln(2\pi n).$$

Hence, with $q = k/n$ and the binary **entropy**

$$H(q) = -q\ln q - (1-q)\ln(1-q),$$

$$\ln\binom{n}{k} \approx nH(q).$$

The count of a type grows like $e^{nH(q)}$: entropy counts configurations.

Entropy against energy decides the type

Type $q = k/n$

$$\begin{pmatrix} \ln P(K = k) \approx n\,[H(q) - E(q)] \\ E(q) = -q \ln p - (1-q)\ln(1-p) \end{pmatrix}$$

Count times weight, in logs.

Now put count and weight together in logs. Count $e^{nH(q)}$ favours $q = 1/2$; weight $e^{-nE(q)}$ favours $q = 1$.

$H(q) - E(q) = -D_{\mathrm{KL}}(q\|p)$: the winner is $q = p$, others decay like e^{-nD}.

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Method of types: probability concentrates on the type matching p, not on one configuration.

Detailed derivation in Further reading

The ten-bit example, type by type

$n = 10$, $p = 0.8$.

k	$P(K = k)$	$D(k/10 \,\|\| \, 0.8)$	e^{-10D}
7	0.2013	0.0282	0.7545
8	0.3020	0	1
9	0.2684	0.0367	0.6929
10	0.1074	0.2231	0.1074

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e^{-nD} is a rate, not a probability.

The same balance is a free energy

Per variable, at $T = 1$

$$\begin{pmatrix} F(q) = E(q) - H(q) \\ \ln P(K = k) \approx -nF(q) \end{pmatrix}$$

Smaller F, more probable type.

The balance $H - E$ has a name in physics: $E - TH$ is the **Gibbs free energy**. Energy pulls toward the lowest-energy configuration, entropy toward the largest count, T sets the exchange rate.

The most probable type has the smallest free energy.

Which q minimizes F, and is the minimum unique? Next page.

The free energy has one minimum, at $q = p$

Claim. $F(q) = D_{\mathrm{KL}}(q\|p) \geq 0$, with equality only at $q = p$, and F is convex. Proof:

$$F(q) = -q \ln p - (1-q)\ln(1-p) + q \ln q + (1-q)\ln(1-q) = q \ln \frac{q}{p} + (1-q) \ln \frac{1-q}{1-p}.$$

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So F falls to its single minimum 0 at $q = p$ and rises on both sides; every other type pays a strictly positive free energy.

One type far from p is exponentially rare

How rare is a type away from p? Write $P(K = k) = c_n(q)\, e^{-nF(q)}$ with $F = D_{\mathrm{KL}}(q\|p)$ and $c_n(q) \le 1$. Pick a tolerance $\delta = 0.1$: a **far type** has $|q - p| > \delta$.

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F is convex with its only zero at p, so outside $[p - \delta, p + \delta]$ it is smallest at the two ends:

$$F(q) \ge F_\delta = \min\{F(p - \delta), F(p + \delta)\} > 0.$$

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Hence a far type has probability at most e^{-nF_δ}. For $p = 0.8$: $F(0.7) = 0.0282$ and $F(0.9) = 0.0367$, so $F_\delta = 0.0282$.

Add up all the far types

One far type is rare; how many are there? Only $n + 1$ types exist, $k = 0, 1, \ldots, n$. Adding the bound over all far types:

$$P(|K/n - p| > \delta) \leq (n + 1)\, e^{-nF_\delta}.$$

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At $n = 100$ the bound is $101\, e^{-2.82} = 6.04$, larger than 1 and useless. At $n = 1000$ it is $1001\, e^{-28.2} = 5.9 \times 10^{-10}$.

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The exponential decay beats the polynomial factor $n+1$ once n is large. Where, then, does the probability lie?

Deriving the method of types

For every tolerance $\delta > 0$, the probability of a type farther than δ from p tends to 0 as n grows.

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The most probable single configuration, $q = 1$, is itself a far type and is exponentially rare.

I. Csiszár, “The method of types,” IEEE Trans. Inform. Theory 44(6):2505–2523, 1998; Cover and Thomas, ch. 11.

From the global minimum to step-by-step choices

Two ways to pick

$$\begin{pmatrix} T \to 0 : \text{ global} \\ \text{decoder: local} \end{pmatrix}$$

Do they agree?

At $T \to 0$ the model concentrates on its minimum-energy configuration, compared against all configurations at once.

A text decoder cannot afford that: it picks the most probable **next** token, one step at a time.

Is the step-by-step winner the global winner? A two-token example decides.

Local greed and global optimization differ

Two-token example

$$\begin{pmatrix} P(A) = 0.6,\ P(B) = 0.4 \\ \max_y P(y \mid A) = 0.5 \\ \max_y P(y \mid B) = 0.99 \end{pmatrix}$$

Best full sequence: compare products.

Greedy first choice: A, best continuation $0.6 \times 0.5 = 0.30$.

Starting with B: $0.4 \times 0.99 = 0.396$, larger.

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Step-by-step greed and the global minimum-energy configuration are different claims. Section III turns to models in which many local choices interact.

Break

10 minutes

Local preferences and global behavior

1. Evaluate predictions
2. Measure information
3. Model with energy

Section II ended with interacting local choices. **Same graph language:** circles and squares, now described by energies.

New difficulty: preferences can conflict around a cycle. A grid can support many different configurations.

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Lecture 4 asks: can measured averages guide the choice of local functions? Maximum entropy supplies one modeling principle.

A spin is a binary variable

Two equivalent encodings

$$\begin{pmatrix} x_i \in \{0,1\} \\ s_i = 2x_i - 1 \in \{-1,+1\} \end{pmatrix}$$

Here i identifies a site.

We build the model from its smallest part. A spin is a two-state variable; we use signs because products express agreement conveniently.

If $s_i = s_j$, then $s_i s_j = +1$. If they differ, $s_i s_j = -1$.

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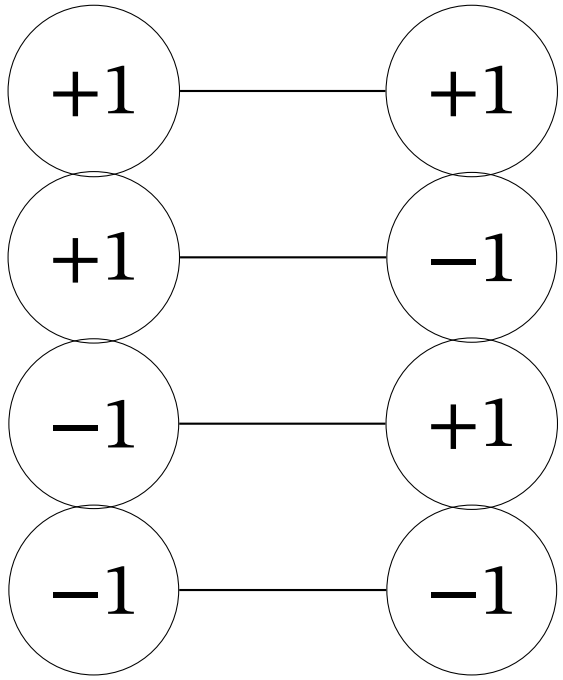
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If $s_i = s_j$, then $s_i s_j = +1$. If they differ, $s_i s_j = -1$.

The model can describe a mathematical binary system without assuming that it is a literal collection of magnetic particles.

A coupling expresses pair preference

Pair energy: $-Js_is_j$

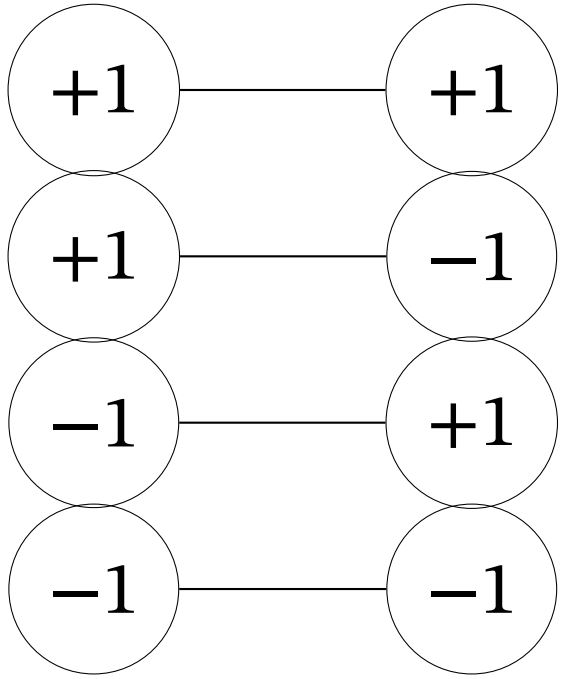


Spins interact in pairs. The coefficient $J \in \mathbb{R}$ is a **coupling**; it contributes $-Js_is_j$ to energy.

If $J > 0$, equal signs give lower energy. If $J < 0$, unequal signs give lower energy.

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The corresponding positive **local function** is $f_{ij}(s_i, s_j) = e^{Js_is_j/T}$.

A field expresses individual preference

Unary energy $-hs_i$

$$\begin{pmatrix} s_i = +1 : & -h \\ s_i = -1 : & +h \end{pmatrix}$$

The field is the real number h.

A site can also have its own preference. For $h > 0$ the unary term favors $+1$; for $h < 0$, -1.

The model combines unary terms with pair couplings.

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A field pushes each spin toward one sign. Pushed hard toward $+1$, both spins almost always agree, but no longer because of each other: the next board shows the **correlation shrinking while J stays fixed**.

Where the table comes from

Two spins, $J = \ln 2/2$, $T = 1$, equal fields h.

The energy is $E(s_1, s_2) = -Js_1s_2 - hs_1 - hs_2$, so the weight factorizes: $e^{-E} = e^{Js_1s_2}\, e^{h(s_1+s_2)}$.

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With $J = \ln 2/2$, $e^{Js_1s_2} = \sqrt{2}$ if the spins agree, $1/\sqrt{2}$ if they differ. Entries: $(-1,-1)$: $\sqrt{2}\, e^{-2h}$; $(+1,+1)$: $\sqrt{2}\, e^{2h}$; mixed signs: $(1/\sqrt{2})e^{0} = 1/\sqrt{2}$.

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Multiplying every entry by $\sqrt{2}$ changes no probability (the normalizer absorbs it) and gives $g = \begin{pmatrix} 2e^{-2h} & 1 \\ 1 & 2e^{2h} \end{pmatrix}$ (order $-1, +1$).

A strong field hides the coupling

Same two spins; $\mathrm{Cov} = \mathbb{E}[s_1 s_2] - \mathbb{E}[s_1]\mathbb{E}[s_2]$**.**

h	Z	$\mathbb{E}[s_i]$	$\mathbb{E}[s_1 s_2]$	$\mathrm{Cov}(s_1, s_2)$
0	6	0	0.3333	0.3333
1	17.0488	0.8509	0.7654	0.0413
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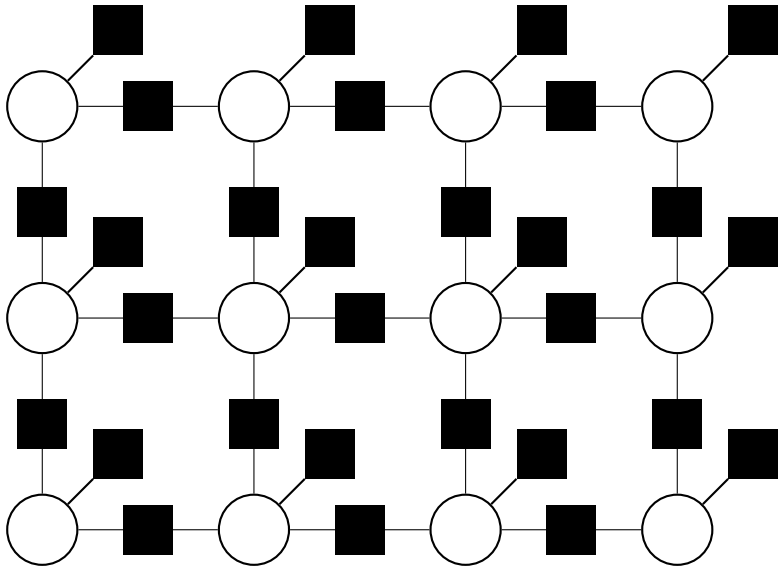
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At $h = 0$ the spins agree only through J: covariance $1/3$.

At $h = 2$ both are $+1$ with probability 0.98: they agree almost always, yet the covariance is 0.001. The agreement comes from the field. J **never changed:** correlation and MI depend on J and h together.

Local energies define a Boltzmann machine



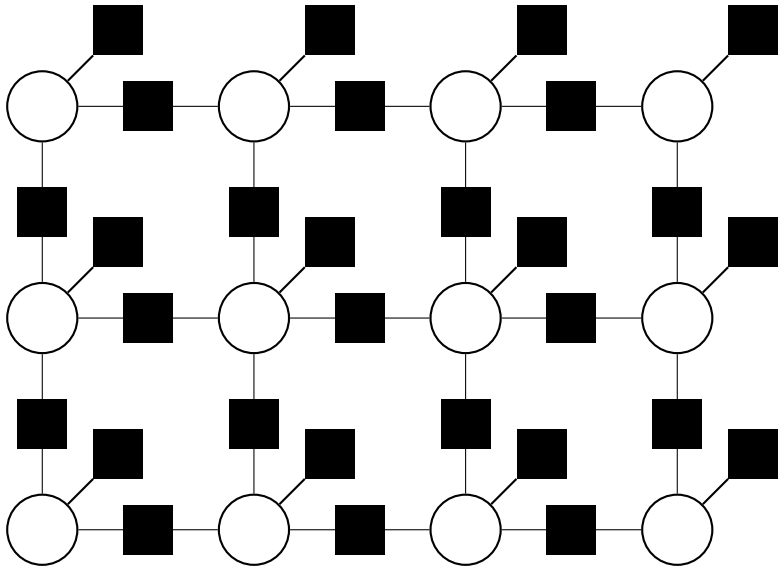
Pair and unary local functions.

Assemble them on a graph. $\mathcal{V}$: unit indices. $\mathcal{E}$: connected pairs, listed once. Off-grid squares: unary functions.

With couplings J_{ij} and fields h_i, define

$$E(s) = -\sum_{\{i,j\}\in\mathcal{E}} J_{ij} s_i s_j - \sum_{i\in\mathcal{V}} h_i s_i.$$

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Normalize: $p_T(s) = e^{-E(s)/T}/Z_T$. This is the **Boltzmann machine**; on a lattice with fixed J, h physics calls it the Ising model.

Energy of the all-plus grid

3×3 grid, $J = 1$ on every neighbour pair, no field.

The grid has 12 neighbour pairs: 6 horizontal and 6 vertical. The energy is $E(s) = -$(sum of the 12 products $s_i s_j$).

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So $E = -12$, the lowest energy possible: no pair disagrees.

Flip the centre spin

Start from the all-plus grid and set the centre spin to -1.

The centre has 4 neighbours. Each of those 4 products turns from $+1$ into -1; the other 8 products stay $+1$.

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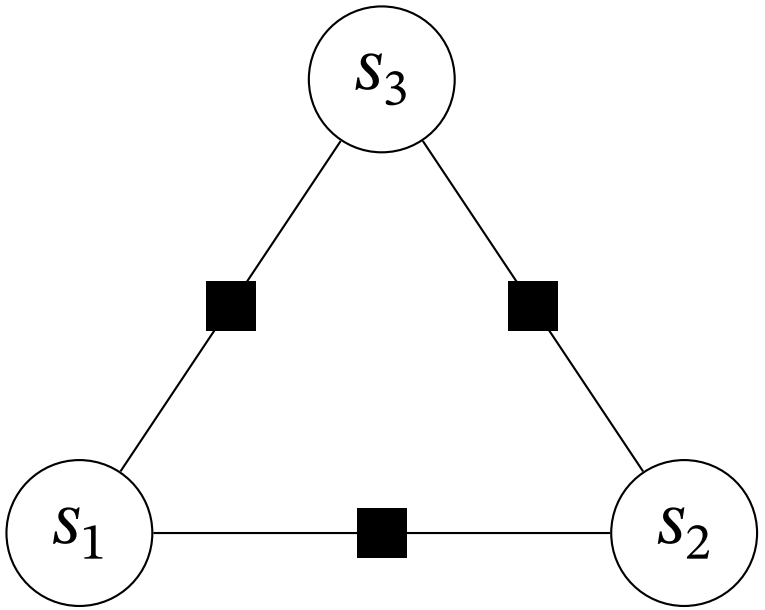
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$p(s) = e^{-E(s)}/Z$ at $T = 1$. In the ratio $p(\text{flipped})/p(\text{all plus}) = e^4/e^{12} = e^{-8} = 0.000335$ the **normalizer** Z cancels. An absolute probability such as $p(\text{all plus}) = e^{12}/Z$ needs $Z = \sum_s e^{-E(s)}$ over all $2^9 = 512$ configurations: the whole sum.

A cycle can contain conflicting preferences

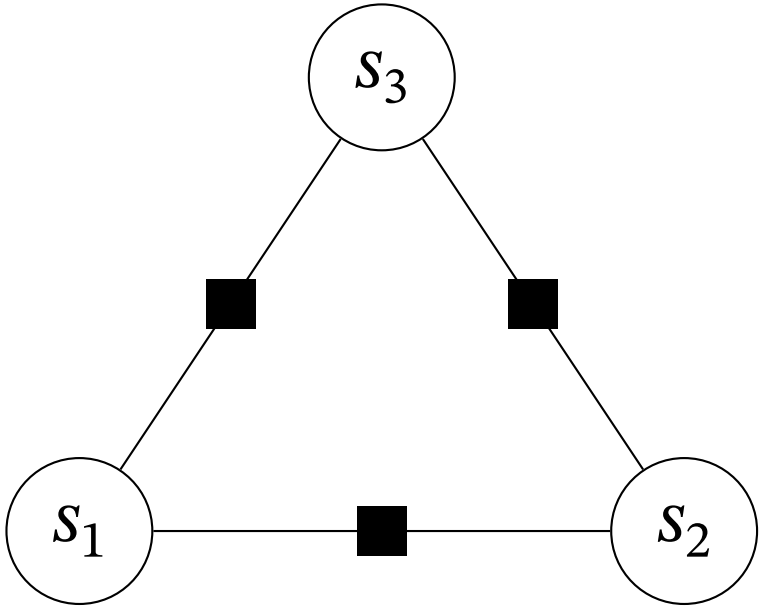


Original triangle: all $J = -1$

On the grid every preference could hold at once. On this triangle it cannot: every pair prefers opposite signs. Start with $s_1 = +1$ and satisfy two edges.

Activity 5. Change one coupling to $+1$. Can all three preferences now hold?

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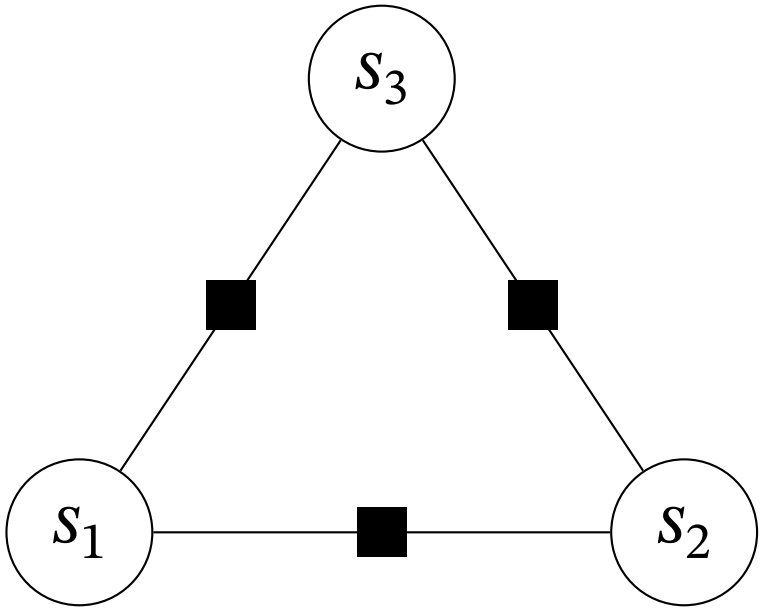
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Your explanation.

Conflict changes the energy spectrum



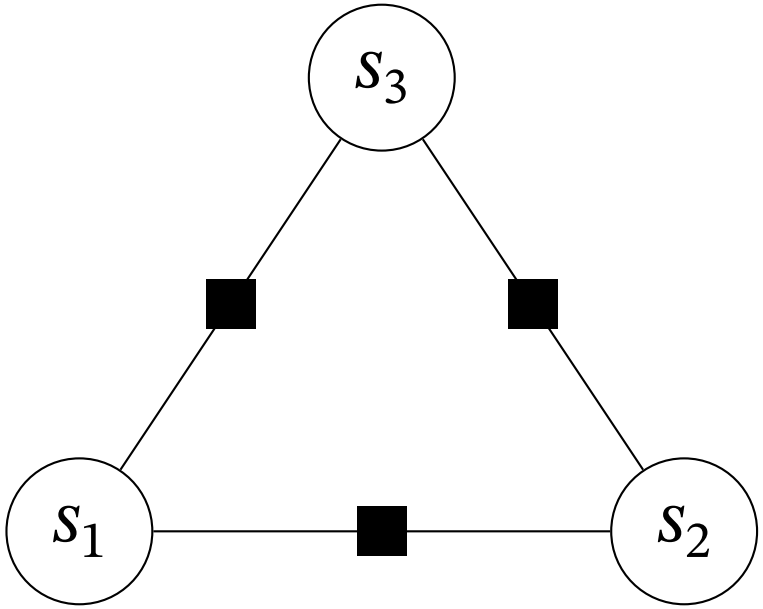
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Return to the original triangle. With three couplings $J = -1$ and zero fields, energy is

$$s_1 s_2 + s_2 s_3 + s_3 s_1.$$

Six configurations have energy -1. The two all-equal configurations have energy 3.

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The ground state is not unique. Both energy and the number of states at each energy affect the distribution.

Counting states at each energy

Three spins form a triangle with $J = -1$, zero fields and $T > 0$.

Configuration type	Number	Energy	Total weight
All signs equal	2	3	$2e^{-3/T}$
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Each **minimum** receives probability 0.1657. One state's probability differs from the group's probability.

Zero mean can hide strong structure

A mixture of two configurations

$$\begin{pmatrix}(+1,+1,+1) : 1/2\\(-1,-1,-1) : 1/2\end{pmatrix}$$

Every spin has mean zero.

Later we fit such models from measured averages. Which averages? Averaging each spin separately gives $\mathbb{E}[s_i] = 0$.

Yet each pair always agrees, so $\mathbb{E}[s_i s_j] = 1$.

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A list of individual means does not describe the **joint distribution**. Pair statistics reveal information the means omit.

The same grid supports many states

$+1$	$+1$	$+1$	$+1$
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All sites agree.

-1	-1	-1	-1
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Back on the grid, with positive nearest-neighbor couplings, an edge joining unlike signs costs more energy than an aligned edge.

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At positive **temperature**, higher-energy configurations still receive probability. A finite grid has no thermodynamic singularity.

A positive binary table has a coupling form

The four log values

$$\begin{pmatrix} \ln f(-1,-1) \\ \ln f(-1,+1) \\ \ln f(+1,-1) \\ \ln f(+1,+1) \end{pmatrix}$$

Four values determine four coefficients.

Week 2's tables also fit the coupling form. For positive f, write

$$\ln f(s_i, s_j) = c + J s_i s_j + a s_i + b s_j.$$

Four unknowns c, J, a, b, four equations, one per row: exactly one solution.

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So every positive two-spin table is a coupling. A three-spin table has eight values and also needs $s_i s_j s_k$: pairs alone do not suffice.

Detailed derivation in Further reading

One table, one exact reconstruction

Original table f_{23}

$$\begin{pmatrix} 3 & 1 \\ 1 & 4 \end{pmatrix}$$

Rows and columns use $x = 0, 1$.

For $s = 2x - 1$, the coefficients are $c = J = \frac{1}{4} \ln 12$ and $a = b = \frac{1}{4} \ln(4/3)$.

At $s_i = s_j = -1$, the log value is $c + J - a - b = \ln 3$.

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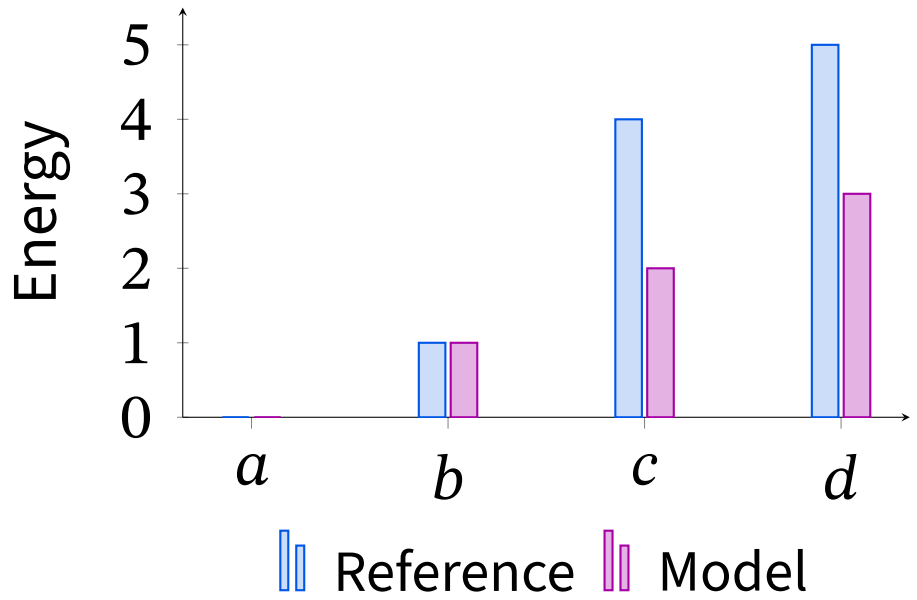
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The other three rows reconstruct in the same way. So far every energy was given. What if energies are learned from some states only?

A perfect fit on observed states



Teaching model:

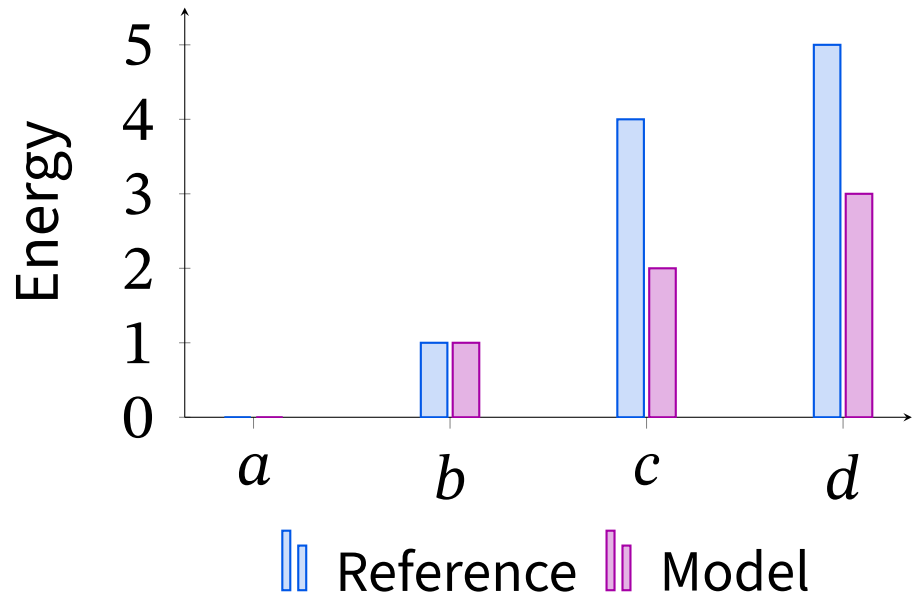
$x \in \mathcal{X} = \{a, b, c, d\}$ labels four configurations. Energy is dimensionless.

$E_{\mathrm{ref}} = (0, 1, 4, 5).$

$E_{\mathrm{model}} = (0, 1, 2, 3).$

Constructed values, not materials data.

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Observed a, b**:** zero error. **Unobserved** c, d**:** energy too low by 2.

What happens to probabilities?

The normalizer includes unobserved states

At $T = 1$, compute $g(x) = e^{-E(x)}$ and $Z = \sum_x g(x)$.

	a	b	c	d
g_{ref}	1	0.3679	0.0183	0.0067
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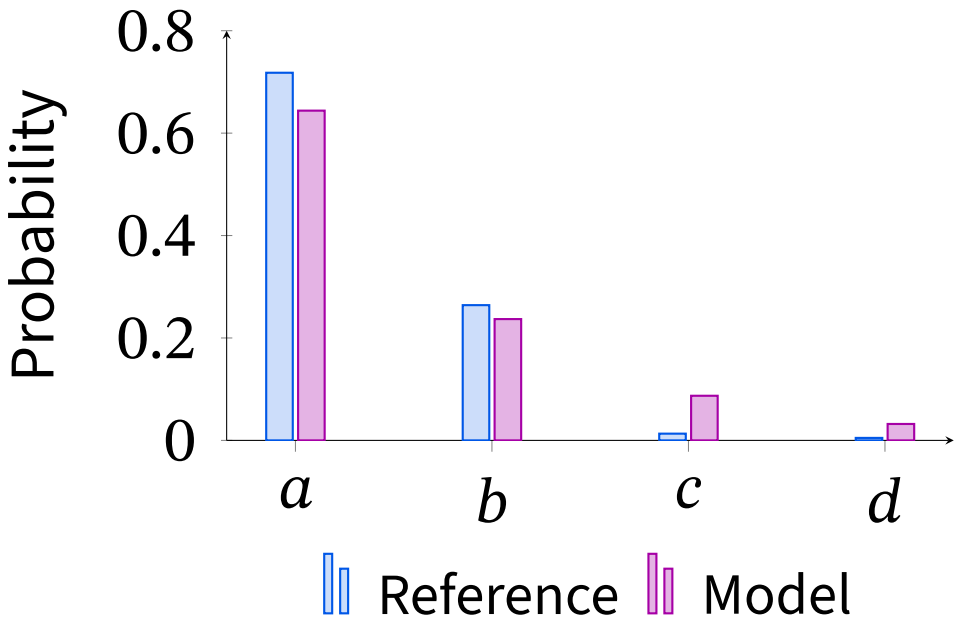
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a has the same energy in both, yet $p_{\text{ref}}(a) = 1/1.3929 = 0.7179$ and $p_{\text{model}}(a) = 1/1.5530 = 0.6439$: extra weight on c, d enlarges Z, so **unseen-state errors leak into seen states**.

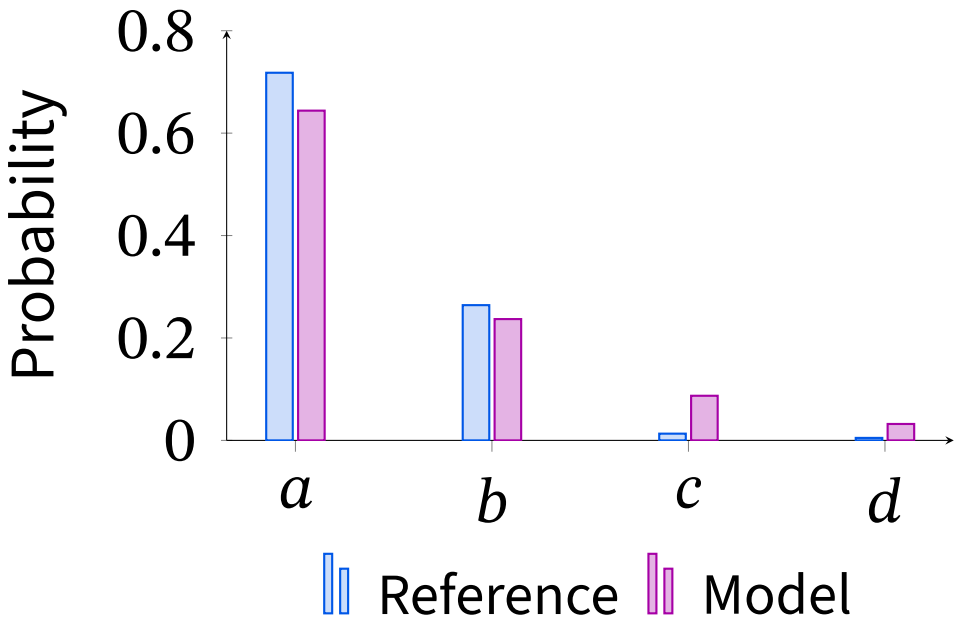
Excess probability on unseen states



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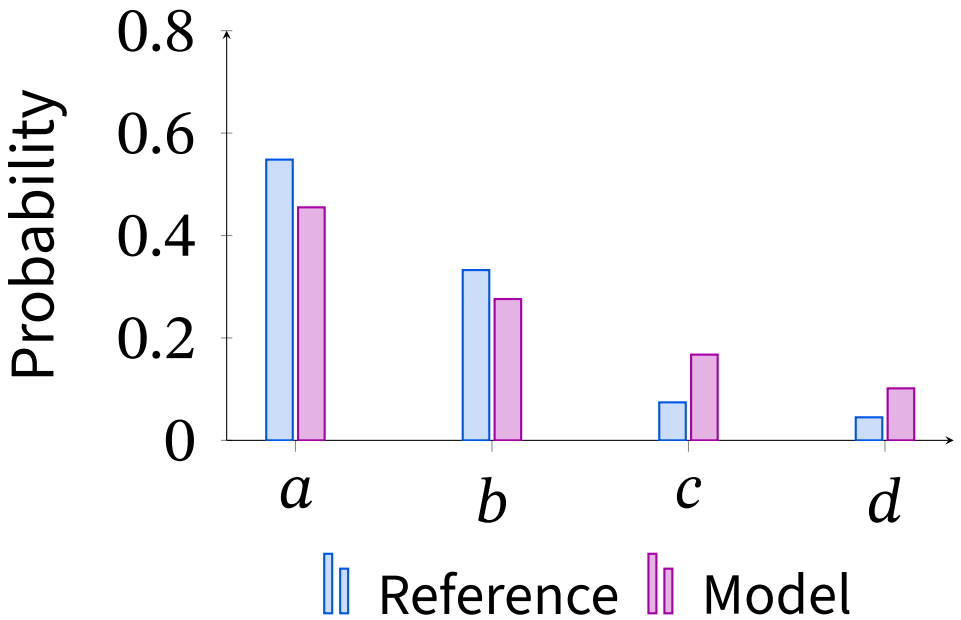


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This is **6.6 times the reference probability**. The training energy error did not reveal it.

The same errors at a higher temperature



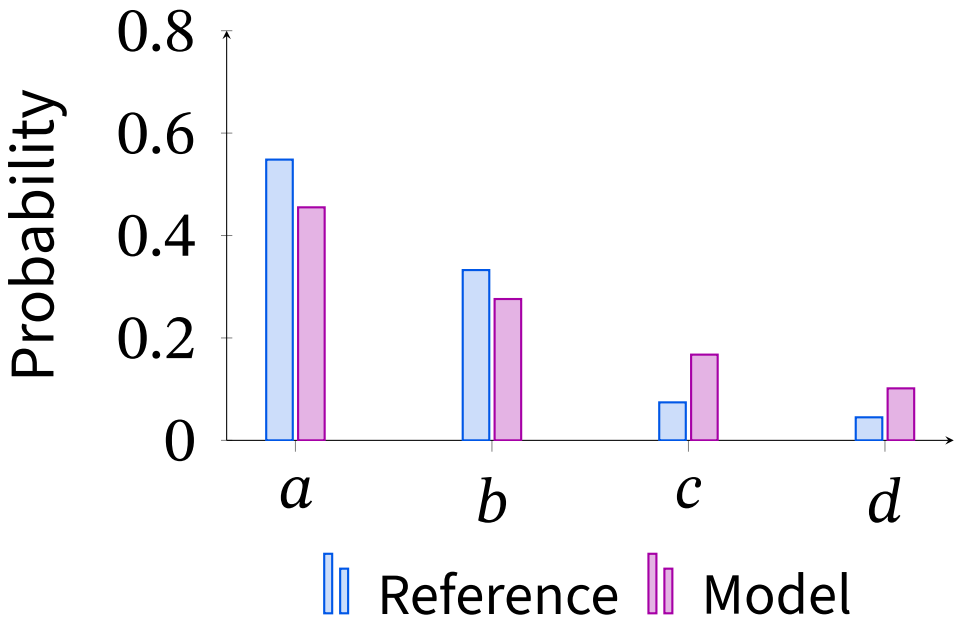
Keep both energies fixed. Set $T = 2$ and normalize $g_T(x) = e^{-E(x)/2}$.

$Z_{\text{ref}} = 1.8240, Z_{\text{model}} = 2.1975.$

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The probability excess rises from 0.1012 to 0.1497. **Identical energy errors can have different statistical effects.**

Which energy errors change probabilities?

A different model

$$\big(E_{\text{new}}(x) = E_{\text{ref}}(x) + 2\big)$$

Add the same amount to every state.

Your turn: would this new model change the normalized probabilities at a fixed $T > 0$?

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Where this lecture stopped

Covered in Lecture 3

$$\begin{pmatrix} \text{I. Evaluating predictions} \\ \text{II. Information and energy} \\ \text{III. The Boltzmann machine} \end{pmatrix}$$

Section III continues in Lecture 4.

We ended on the learned-energy example: a fit can match every observed energy and still misplace probability.

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Then sampling: with n spins the normalizer has 2^n terms, so averages must be estimated. The Further reading pages that follow are optional.

Where this lecture fits

Week 1 Specify local relationships

Week 2 Compute exact marginals on trees

Week 3 Evaluate and explain distributions

Week 4 Estimate difficult averages by sampling

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Week 4 Estimate difficult averages by sampling

Lecture 4: maximum entropy and moment matching, then sampling to estimate the averages needed to evaluate and fit large graph models.

Office hours: Yuwen Huang, Tuesday 11:00–12:00, W4 L4 422.

Closing a region precisely

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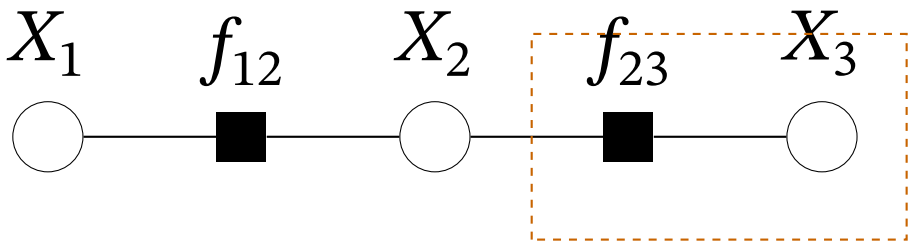
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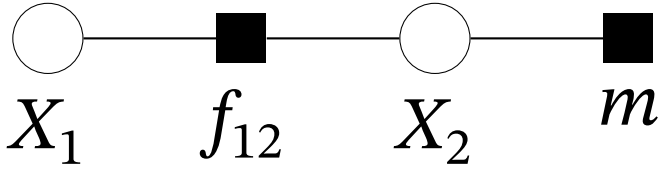
$$f_{\mathcal{R}}(x_{\mathcal{B}}) = \sum_{x_{\mathcal{U}}} \prod_{f \in \mathcal{R}} f(x_{\mathcal{N}(f)}).$$

Variables in $\mathcal{U}$ disappear. The partition function and **marginals** of retained variables stay unchanged. The global function is now defined on a smaller domain.

Opening exposes a representation



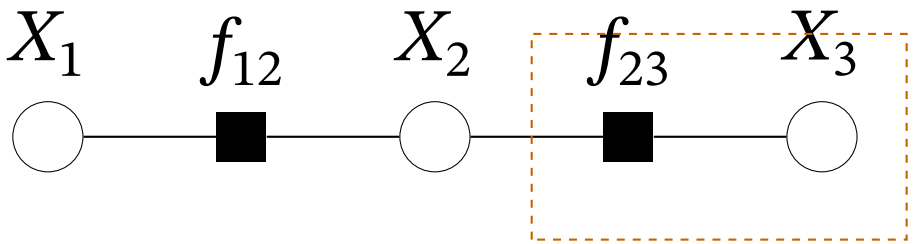
Sum over X_3 inside the box.



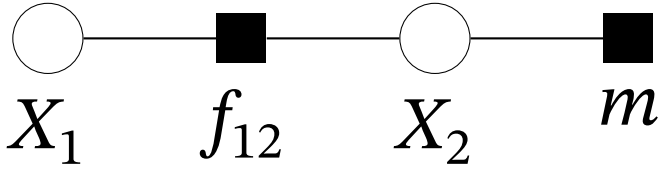
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Enumerating N assignments with r **local functions** takes $N(r-1)$ product multiplications. Reusing local sums can reduce this work.

Return to the lecture example
Further reading

Entropy from local marginal distributions

For positive **local functions**, $\ln p(x) = \sum_f \ln f(x_{\mathcal{N}(f)}) - \ln Z$. Take the expected negative log under p.

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Hence $H(p) = \ln Z - \sum_f \mathbb{E}_p[\ln f]$. Computing those **marginals** has a separate inference cost. The identity alone does not make inference tractable.

Return to the lecture example

Further reading

KL divergence and support

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No division by $p(y)$ is needed outside its support. All sums in this argument have an explicit domain.

Return to the lecture example

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Equality in the tangent bound requires $q(y)/p(y) = 1$ at every $y \in \mathcal{S}_p$.

These **probabilities** already sum to one. Therefore q cannot have positive mass outside $\mathcal{S}_p$, and equality holds exactly for $p = q$.

For $p = (1, 0)$ and $q = (1/2, 1/2)$, $D(p\|q) = \ln 2$ while $D(q\|p) = +\infty$. The two directions need not agree.

The log-sum inequality

For $a_i, b_i > 0$,

$$\sum_i a_i \ln \frac{a_i}{b_i} \geq \left(\sum_i a_i\right) \ln \frac{\sum_i a_i}{\sum_i b_i}.$$

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Equality holds exactly when all ratios a_i/b_i are equal.

KL divergence is jointly convex

For each y apply the log-sum inequality to $a = (\lambda p_1(y), (1-\lambda)p_2(y))$ and $b = (\lambda q_1(y), (1-\lambda)q_2(y))$, then sum over y.

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$$D(\lambda p_1 + (1-\lambda)p_2 \,\|\, \lambda q_1 + (1-\lambda)q_2) \leq \lambda D(p_1 \| q_1) + (1-\lambda) D(p_2 \| q_2).$$

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Convexity in the pair contains convexity in q alone. Since $H(p) = \ln K - D(p\|u)$, entropy is **concave**, so the maximum-entropy problems of Section III have unique solutions.

Return to the lecture example

Data processing from a chain rule

Define $I(X;Y \mid W) = \mathbb{E}_W[D(p_{XY|W} \| p_{X|W} p_{Y|W})]$. Expanding logs gives $I(X;Y,Z) = I(X;Y) + I(X;Z \mid Y)$.

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For $X \to Y \to Z$, conditional independence gives $I(X;Z \mid Y) = 0$. The other expansion gives

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$$I(X;Y) = I(X;Z) + I(X;Y \mid Z) \geq I(X;Z).$$

The reverse **factorization** also gives $I(X;Z) \leq I(Y;Z)$. Condition only on events of positive probability.

Return to the lecture example

Why the average entropy decreases

An equivalent expression is

$$I(X;Y) = \sum_x p(x) D_{\mathrm{KL}}(p_{Y|x} \| p_Y) \geq 0.$$

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Individual conditional entropies can exceed $H(Y)$. The theorem bounds the average, not every particular observation.

Return to the lecture example

The probability of a type

Type k has probability $\binom{n}{k} p^k (1-p)^{n-k}$. With $q = k/n$, Stirling gives $\ln \binom{n}{k} = nH(q) - \frac{1}{2}\ln(2\pi nq(1-q)) + O(1/n)$.

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$$\frac{1}{n}\ln P(K = k) = -D_{\mathrm{KL}}(q\|p) + O\!\left(\frac{\ln n}{n}\right).$$

Check: $\ln \binom{100}{80} = 47.731$ exactly, 47.735 by Stirling. The rate $D_{\mathrm{KL}}(q\|p)$ is the large-deviation exponent of the empirical frequency.

Return to the lecture example

Finite changes in temperature

Hold the energy fixed. For $0 < T_1 < T_2$, define $\beta_i = 1/T_i$, $U_i = \mathbb{E}_{p_{T_i}}[E]$, $H_i = H(p_{T_i})$, $\Delta U = U_2 - U_1$, and $\Delta H = H_2 - H_1$.

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Expand KL in both directions:

$$0 \leq D(p_{T_1} \| p_{T_2}) = \Delta H - \beta_2 \Delta U,$$

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Therefore $\beta_2 \Delta U \leq \Delta H \leq \beta_1 \Delta U$. Since $\beta_1 > \beta_2 > 0$, both changes are nonnegative. Strictness requires a nonconstant energy.

Temperature increases entropy here

Hold a finite energy function fixed. Define $\beta = 1/T$ and $A(\beta) = \ln \sum_x e^{-\beta E(x)}$. Differentiation gives $A^{'} = -U$ and $A^{''} = \mathrm{Var}(E)$.

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Using $d\beta/dT = -1/T^2$ and $H = U/T + A$,

$$\frac{dU}{dT} = \frac{\mathrm{Var}(E)}{T^2}, \qquad \frac{dH}{dT} = \frac{\mathrm{Var}(E)}{T^3}.$$

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$$\frac{dU}{dT} = \frac{\mathrm{Var}(E)}{T^2}, \qquad \frac{dH}{dT} = \frac{\mathrm{Var}(E)}{T^3}.$$

Both are nonnegative for $T > 0$, and positive unless energy is constant on the support. Changing E at the same time is outside this result.

The limit keeps every minimum

Let $E_* = \min_x E(x)$ and $\mathcal{A} = \{x : E(x) = E_*\}$. Subtract this **minimum**:

$$p_T(x) = \frac{e^{-[E(x)-E_*]/T}}{\sum_y e^{-[E(y)-E_*]/T}}.$$

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The limiting distribution is uniform on $\mathcal{A}$, so its **entropy** is $\ln |\mathcal{A}|$. This statement uses a finite state space with counting measure.

Return to the lecture example

Further reading

Hard constraints remain hard

If some weights vanish, define $\mathcal{S} = \{x : g(x) > 0\}$. Forbidden states remain forbidden for every $T > 0$.

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As $T \to \infty$, $g(x)^{1/T} \to 1$ on $\mathcal{S}$. The limit is uniform on the allowed set, not on all configurations.

If $\mathcal{S}$ is empty, then $Z_T = 0$. No probability distribution exists. Returning a uniform vector would silently change the model.

A rescaled sum approaches a maximum

For real $a_1, \ldots, a_K$ and $m = \max_k a_k$,

$$e^{m/T} \leq \sum_k e^{a_k/T} \leq K e^{m/T}.$$

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Apply $T \ln$ to each term:

$$m \leq T \ln \sum_k e^{a_k/T} \leq m + T \ln K.$$

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$$m \le T \ln \sum_k e^{a_k/T} \le m + T \ln K.$$

The middle expression tends to m. This connects local sum-product elimination to max-product in rescaled log coordinates. An unscaled raw sum does not become a maximum.

Recovering the coupling coefficient

Write $\ell(s_i, s_j) = \ln f(s_i, s_j) = c + Js_is_j + as_i + bs_j$ for a positive binary table.

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$$4J = \ell(-1,-1) + \ell(+1,+1) - \ell(-1,+1) - \ell(+1,-1).$$

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$$4J = \ell(-1,-1) + \ell(+1,+1) - \ell(-1,+1) - \ell(+1,-1).$$

Thus $J = \frac{1}{4} \ln[f_{00} f_{11} / (f_{01} f_{10})]$. If $J = 0$, this **local function** separates into unary functions. Other paths can still create dependence.

Return to the lecture example

Recovering the remaining coefficients

Average the four log entries to obtain c. Average their products with s_i to obtain a, and with s_j to obtain b.

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Recovering the remaining coefficients

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The functions $1, s_i, s_j, s_i s_j$ are orthogonal under the uniform average on $\{-1, +1\}^2$. Each has squared average 1.

They form a basis of all functions on four states, so the coefficients are unique. Zero table entries require infinite energies rather than this finite representation.

Self-test: find the missing condition

"Low **entropy** means accurate predictions." What is missing?
Your explanation.

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“Low **entropy** means accurate predictions.” What is missing?
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“Pairwise independence implies joint independence.” Give a counterexample.
Your explanation.

Self-test: find the missing condition

“Low **entropy** means accurate predictions.” What is missing?
Your explanation.

“Pairwise independence implies joint independence.” Give a counterexample.
Your explanation.

“The **normalizer** never matters.” Give a counterexample.
Your explanation.

Practice-sheet reference guide: results 1–10

Earlier number	Current explanation
1	Closing a region precisely
2	Opening exposes a representation
3	Retained variables and preserved probabilities
4	Surprise and entropy
5	Entropy from local marginal distributions
6	KL divergence
7	The logarithm tangent inequality
8	Gibbs’ inequality
9	Entropy bounds
10	Cross-entropy

Practice-sheet reference guide: results 11–20

Earlier number	Current explanation
11	Cross-entropy, entropy and KL identity
12	Likelihood and cross-entropy
13	Conditional entropy
14	Mutual information
15	Conditional mutual information
16	Mutual information identities
17	Data processing
18	Energy as a negative log weight
19	Boltzmann distribution
20	Temperature and energy conventions

Practice-sheet reference guide: results 21–29

Use the revised statements and their explicit conditions.

Earlier number	Current explanation
21	Temperature changes and monotonicity
22	Temperature limits and tied minima
23	A rescaled log sum approaches a maximum
24	Binary-table coupling coefficients
25	The Boltzmann machine
26	Lecture 4, moment matching: derivatives of $\ln Z$
27	Lecture 4, maximum entropy at a fixed average energy
28	Lecture 4, $\ln Z$ as the best entropy–energy trade-off
29	Lecture 4, maximum entropy with several averages

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